

**Experience Shapes Early Noun Comprehension from 8-18 Months: The Roles
of Word Frequency and Referent Familiarity**

Lily Zihui Zhu^a, Andrei Amatuni^b, Shannon Egan-Dailey^c, Hallie Garrison^c, Evgenii
Kalenkovich^a, Sharath Koorathota^d, Lillianna Righter^a, Shaelise Tor^d, and & Erika
Bergelson^{a,c}

^a Harvard University

^b University of Texas-Austin

^c Duke University

^d University of Rochester

Author Note

Other than the first and last authors, all authors are listed alphabetically according
to their last names.

Correspondence concerning this article should be addressed to Lily Zihui Zhu, 33
Kirkland St, Cambridge, MA 02138. E-mail: lilyzhu@fas.harvard.edu

Abstract

We investigated how experience with particular nouns and referents influences infant word comprehension by combining home recordings with in-lab eyetracking. In Study 1, infants (n=44) participated in a yearlong longitudinal study which included monthly home recordings (daylong audio and hourlong video, 6-17mo.) and hand-tailored eyetracking experiments (every two months, 8-18mo.). Word comprehension was operationalized as how much more infants looked at named images than distractors. Each session tested 16 words: 8 held constant across all infants at each timepoint (“generic” common nouns, e.g., “shoe”), and 8 hand-picked for each session from each child’s recent home recordings. Of the hand-picked items, half appeared as images familiar to children from home (e.g., their own ball); the other half and all generic items appeared as prototypical child-friendly (but not individually-tailored) images. Results showed expected improvements with age, and an effect of word frequency (particularly at 12 and 14mo. timepoints) but not referent familiarity. In Study 2 (n=247), cross-sectional controls completed the same eyetracking experiments (i.e., those hand-tailored for Study 1 children) as a stronger test of the roles of frequency and referent familiarity. Here, we did find an effect of referent familiarity. Study 2 infants understood words less well than those in Study 1, particularly when tested with images of referents selected from Study 1 children’s homes. Together, these findings highlight the role of experience in shaping early noun comprehension and suggest that once infants understand words, they readily generalize them to new instances beyond those they are familiar with.

Keywords: word learning, language development, eyetracking, concept, frequency, infancy

Word count: 14711 (main text), 2352 (references)

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1 Introduction

Speakers of a given language come to share a common vocabulary of thousands of frequent words. For the small set of closed-class items like ‘the’, ‘of’, or ‘and’, this is relatively obvious: Words like determiners, prepositions, and conjunctions are language’s glue, and are essentially unavoidable in samples of speech longer than a few seconds. While the learning process for such words is anything but obvious, no learner goes unexposed to the core set of closed-class words in their language and the many varying contexts in which they are used. In contrast, the driver of frequency for open class words like nouns is not grammatical necessity, but lexical and conceptual commonality. That is, common nouns refer to *concepts* that frequently arise across communicative contexts, e.g., ‘dog’ or ‘shoe’ in speech to young children in many communities around the world. On the other hand, concrete words like ‘axolotl’ or ‘joist’ arise rarely in daily conversations, except among herpetologists and carpenters.

This raises a central question for early word learning: How does the frequency and context of exposure to particular nouns shape infants’ acquisition of these words? Word learning involves identifying how wordforms link to specific referents and broader conceptual categories. Two dimensions of word learning experience that can shape early comprehension, especially for concrete nouns, are word frequency (exposure to wordforms) and referent familiarity (exposure to specific referents). For example, the word ‘dog’ and pet dogs likely occur more frequently in children’s home environments than ‘axolotl’ and pet axolotls. Thus, word frequency can serve as a proxy for the amount of word learning input children receive. In addition, because words are typically heard in specific contexts with particular referents, experience with individual referents may influence learning in ways that go beyond frequency alone. For example, a child who hears ‘dog’ co-occurring

with their pet dog at home needs to eventually generalize the word to other, less familiar instances of dogs.

In what follows, we investigate how everyday interactions with particular words and referents connect to early learning. We home in on common concrete nouns, given evidence that infants begin to understand them in their second half-year and expand this knowledge substantially in year two, across socio-cultural and linguistic contexts (e.g., Bergelson, 2020; Fernald et al., 1998; Foushee & Srinivasan, 2024). More concretely, we ask how experience with particular nouns and referents ties to 8-18-month-olds’ abilities to understand words in the lab. To do so, we quantify infants’ experience with individual words and referents based on naturalistic home recordings, and experimentally assess how word frequency and referent familiarity affect infants’ word comprehension in a controlled lab setting.

We first review prior works on the role of word frequency in language development, and work on how referent familiarity can shed light on the scope of early semantic representations. We then describe our approach and present key predictions.

1.1 Word Frequency and Word Learning Outcomes

Existing research has sought to quantify infants’ everyday experiences with common nouns and their referents, in order to establish how, in principle, early linkages among them may occur. One study, which uses overlapping data with the present work, found that 6- and 7-month-old English-learning infants in the U.S. heard roughly 700 noun tokens/day, about half of which co-occurred with a referent that was present and attended to. These 700 tokens spanned about 200 distinct noun types on average. If this estimate is representative, then infants would hear roughly a quarter of a million noun tokens in their first year, around half of which provide some co-occurrence between referent and label (Bergelson et al., 2018).

These kinds of early experiences may provide the fodder for ‘brute force’

associations to jump-start the early lexicon (Meylan & Bergelson, 2022). How precisely this works is a debated topic, with proposals ranging from more top-down processes like hypothesis testing (Trueswell et al., 2013), a reliance on social intent (Tomasello & Akhtar, 1995), and biases in how concepts and words are connected (Waxman & Gelman, 2009), to more bottom-up associative learning mechanisms (Smith & Yu, 2008). For example, under a hypothesis-testing framework (often described as ‘propose-but-verify’), infants generate an initial guess about a word’s meaning based on a single exposure and only revise it if later evidence contradicts that guess (Medina et al., 2011; Trueswell et al., 2013). Particularly clear and unambiguous instances may be crucial for successful learning (Cartmill et al., 2013). In contrast, cross-situational learning theories posit that infants accumulate statistical evidence over multiple exposures, gradually narrowing down the possible meanings by tracking which referents consistently co-occur with which words (Smith & Yu, 2008; Yu & Smith, 2007). Related work also highlight the potential roles for incremental visual category learning paired with “bursty” episodes of word-object pairings (Clerkin & Smith, 2022).

Importantly, despite differences in proposed learning mechanisms, these accounts broadly converge on the prediction that higher word frequency should lead to better word learning outcomes. Whether learning proceeds through gradual associative accrual or is guided by top-down knowledge, increased exposure to words and referents provides more opportunities for learning. Greater input exposure not only strengthens statistical regularities but also increases the likelihood of encountering particularly clear, socially rich, and unambiguous learning contexts.

Prior research has shown that word frequency robustly predicts word learning outcomes, after accounting for other factors such as word class and imageability. These studies typically assess word learning outcomes using the MacArthur–Bates Communicative Development Inventory (CDI, Fenson et al., 1993), a parent-report survey

most of which is a vocabulary checklist, and fit statistical models that include frequency as one of several predictors. Some studies have examined item-level learning outcomes (i.e., whether a child understands or produces a particular word) using such data from large cross-sectional samples (Braginsky et al., 2019; Coffey et al., 2024) or smaller longitudinal samples (Swingley & Humphrey, 2018). Others have modeled age of acquisition, often defined as the age by which 50% of children understand or produce a given word (Bleses et al., 2018; Goodman et al., 2008; Hansen, 2017; Hao et al., 2015; Smolík, 2019). Importantly, frequency effects have been observed across multiple languages beyond English, including Mandarin, Norwegian, Czech, Dutch, and Danish (Bleses et al., 2018; Hansen, 2017; Hao et al., 2015; Smolík, 2019; Verhagen et al., 2022). Moreover, Braginsky et al. (2019) modeled acquisition trajectories for early-learned words across 10 languages and found consistent frequency effects. Together, these studies suggest that frequency consistently emerges as a strong predictor of early lexical development well beyond the highly studied English learning context.

Despite the strength of this evidence, most studies linking word frequency to word learning outcomes rely on parent vocabulary checklists rather than objective assessments of comprehension in controlled experimental settings. In addition, most relevant studies use frequency estimates aggregated across many children or families from relatively similar samples (e.g., based on CHILDES), rather than frequency estimates derived from an individual child’s input. To our knowledge, little prior work has directly tested whether child-specific word frequency predicts infants’ real-time comprehension of those same words.

One paradigm well suited for measuring early word comprehension in real time is the “looking while listening” eyetracking paradigm, which has been widely used to assess children’s word comprehension in the lab (Fernald et al., 1998). In this task, children view two images while hearing a label for one of them, and their gaze patterns are tracked, with various measures of “target looking” computed to quantify word comprehension. This

paradigm is useful for asking whether infants in a given age range recognize a set of familiar, imageable words, and for testing how features of the auditory signal, visual display, or competitor structure affect online word comprehension. However, traditional looking-while-listening designs are not typically optimized to measure frequency effects in early word learning. Test words are usually fixed across participants and selected to be highly imageable and likely known to many children in the target age range. These choices increase the likelihood of detecting reliable recognition of the selected word set by the participant group, but they also constrain the range of frequency variation among test items and the “fit” of particular words or referents with a given child. Moreover, especially early in development, infants’ exposure to individual words may measurably vary across households, and a fixed stimulus set may not instantiate the same frequency contrast for every child. Moreover, especially early in development, infants’ exposure to individual words may vary substantially across households, and a fixed stimulus set may not instantiate the same frequency contrast for every child.

The present study addresses this gap in two ways. First, rather than relying on aggregate frequency estimates, we estimate word frequency separately for each infant based on their own recent home recordings (with comparisons to other approaches as well). Second, rather than relying on parent-report vocabulary checklists, we use a hand-tailored looking-while-listening task to assess infants’ real-time comprehension of the same words whose frequency was estimated from their own input. Prior work has shown that adults (e.g., Dahan et al., 2001; Magnuson et al., 2007) and toddlers (Potter & Lew-Williams, 2024) process higher-frequency words more efficiently or accurately in such tasks, but these studies typically involve words that participants already know, and thus focus on how existing representations are accessed in real-time rather than measuring word learning outcomes *per se*. In contrast, our study uses the looking-while-listening paradigm not to study processing efficiency, but to assess whether children have learned specific words. By combining word frequency estimates based on naturalistic home recordings with

individually tailored experimental measures of real-time comprehension, we can test for converging evidence that early word frequency predicts children’s word knowledge.

1.2 Referent Familiarity and Early Semantic Representations

While much prior work has focused on word frequency, another important question concerns the nature of the resulting lexical representations. Infants begin to understand common nouns like ‘hand’ and ‘daddy’ by 6-9 months (Bergelson & Swingley, 2012; Foushee & Srinivasan, 2024; Kartushina & Mayor, 2019; Tincoff & Jusczyk, 1999, 2012), but what is it that they have learned about these words and how specific are these early word representations? Are infants mapping words to specific, individual referents in their environment, or to more abstract conceptual categories?

Researchers have investigated the scope of early lexical representation by testing how infants extend words to different types of exemplars. A narrow scope suggests infants map words to specific referents, while a broad scope suggests infants map words to broad conceptual categories and extend words to a variety of category members. Views on how this ability develops typically fall into two camps. On the narrow-to-broad account, infants initially associate words with individual referents and later generalize them to categories as they encounter more exemplars (Hirsh-Pasek et al., 2004; Smith, 2000). On this view, category structure emerges inductively through experience. For example, children might eventually infer that objects labeled ‘ball’ tend to be round (Smith et al., 2002). In contrast, the broad-to-narrow view holds that infants begin with a bias to treat words as referring to categories of similar objects and fine-tune the categories over development (Csibra & Shamsudheen, 2015; Waxman & Gelman, 2009). Complicating this dichotomy, it is possible that infants entertain both scopes simultaneously. Words can be used to label individuals or categories, and syntax often provides cues to this distinction. English syntax, for example, typically marks common nouns with articles (denoting *category* members, e.g., ‘a dog’) and proper nouns without them (denoting *specific individuals*, e.g., ‘Sarah’).

A few recent experiments test these possibilities. Campbell and Hall (2022) report an interesting set of experiments that provide evidence that infants have both individual- and category-level scopes for their early lexical entries. In a series of eyetracking experiments, they found that 6-month-olds restrict the term ‘Mommy’ to their mother (and not other familiar women), i.e., giving it individual scope, but that 6- and 9-month-olds appropriately extend ‘hand’ and ‘ball’ to multiple category members, i.e., giving it categorical-scope. At 12-15 months, infants can apply both kinds of scope to the same entity, e.g., knowing that both ‘Fluffy’ and ‘dog’ apply to their pet specifically.

Another recent study found that typicality had little impact on 14-18-month-old infants’ ability to recognize words for familiar animal categories (Weaver et al., 2024). In their experiment, infants were presented with pairs of images that were either both typical or both atypical exemplars of a familiar animal (dog, cat, bird, or fish). While overall performance improved with age, word comprehension remained comparable across typicality conditions over time. Taken together, these studies suggest that infants map early words both narrowly to specific referents (e.g., specific social partners) and broadly to conceptual categories (e.g., animal species).

Coming from a different angle, a few further studies have queried the role of semantic similarity in early word learning (Arias-Trejo & Plunkett, 2010, 2013; Bergelson & Aslin, 2017a; Bergelson & Aslin, 2017b; Wojcik & Saffran, 2013), similarly to how it has been examined in adults (Huettig & Altmann, 2005). In one case, Bergelson and Aslin (2017b) found that 6-month-olds understood words better when queried in the context of two unrelated images than two related ones. For example, when asked to look at the milk, infants succeeded if the two images on the screen were a glass of milk and a foot, but not if the images were a glass of milk and a glass of juice. In another study with 12-20-month-olds, Bergelson and Aslin (2017a) found that closer to age one, infants have insufficiently precise semantic categories: When shown unrelated image pairs (e.g.,

cookie–nose) infants looked equally at the target image (cookie) whether they heard its correct label (‘cookie’) or a related but incorrect one (‘banana’). By around 20 months, toddlers differentiated these cases, looking more when the word matched the image than when it was related but incorrect. Together, these studies suggest that even our earliest words are not conceptual islands but rather are part of an interconnected conceptual and lexical space (although frequency differentials between presented items too may play a role, Kartushina & Mayor, 2019).

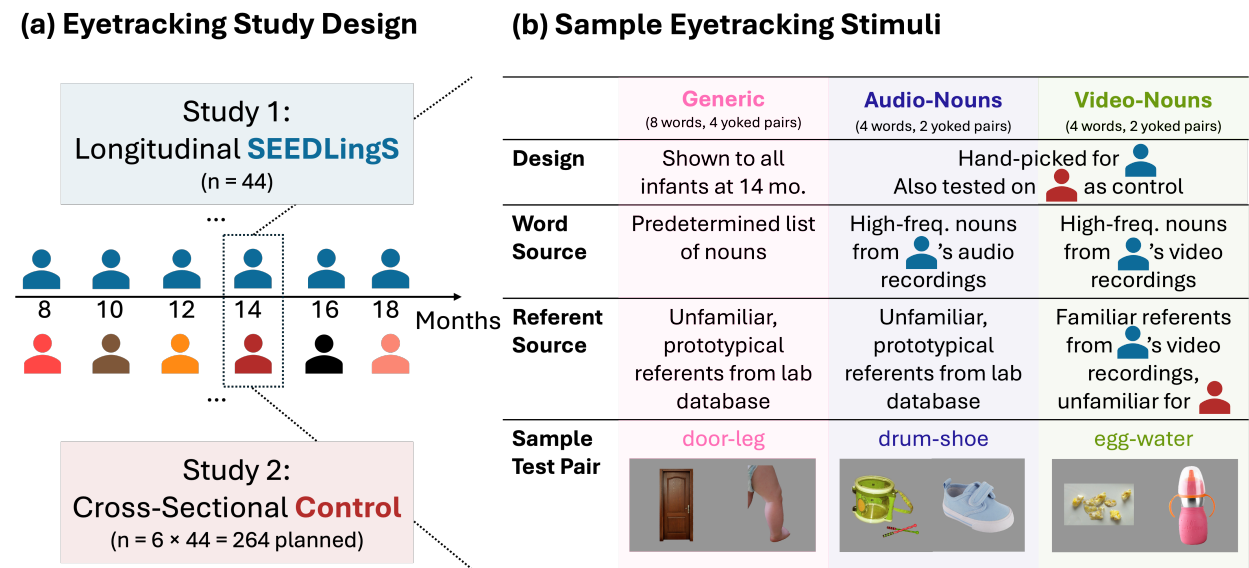
In this paper, we probe the scope of infants’ lexical representations by testing their word knowledge in the context of specific category members that they have variable experience with, and examining how that experience affects word comprehension. If infants’ early representations are narrow, we would expect them to comprehend words more readily when tested on familiar referents. If their representations are broader, we would expect successful comprehension even when tested on unfamiliar category members, though familiar referents may still confer a processing advantage .

Along similar lines, Garrison et al. (2020) investigated whether 12-18-month-old infants recognize words more easily when referents were highly familiar. Infants were presented with pairs of items that were either images of their own referents (e.g., their shoe and toy drum) or a generic, child-friendly version of these items. Older infants linked both the familiar and unfamiliar referents to the target label, but younger infants were only above chance for familiar items. This pattern suggests that early words may be tied to a narrow set of exemplars specific to each infant’s experience. However, these effects were inconsistent across analyses, and no interaction between age and familiarity was observed, leading the authors to conclude that referent familiarity played only a limited role in word comprehension. Still, Garrison et al. (2020) offer suggestive evidence that younger infants may be more sensitive to familiarity. Here, we explore this possibility by extending the age range downward to 8 months and directly linking infants’ real-world experience with

specific referents to their in-lab comprehension of the corresponding words. On a narrow-to-broad account, we predict the largest gap in comprehension between familiar and unfamiliar referents in the youngest infants, with this gap decreasing with age.

1.3 The Present Study

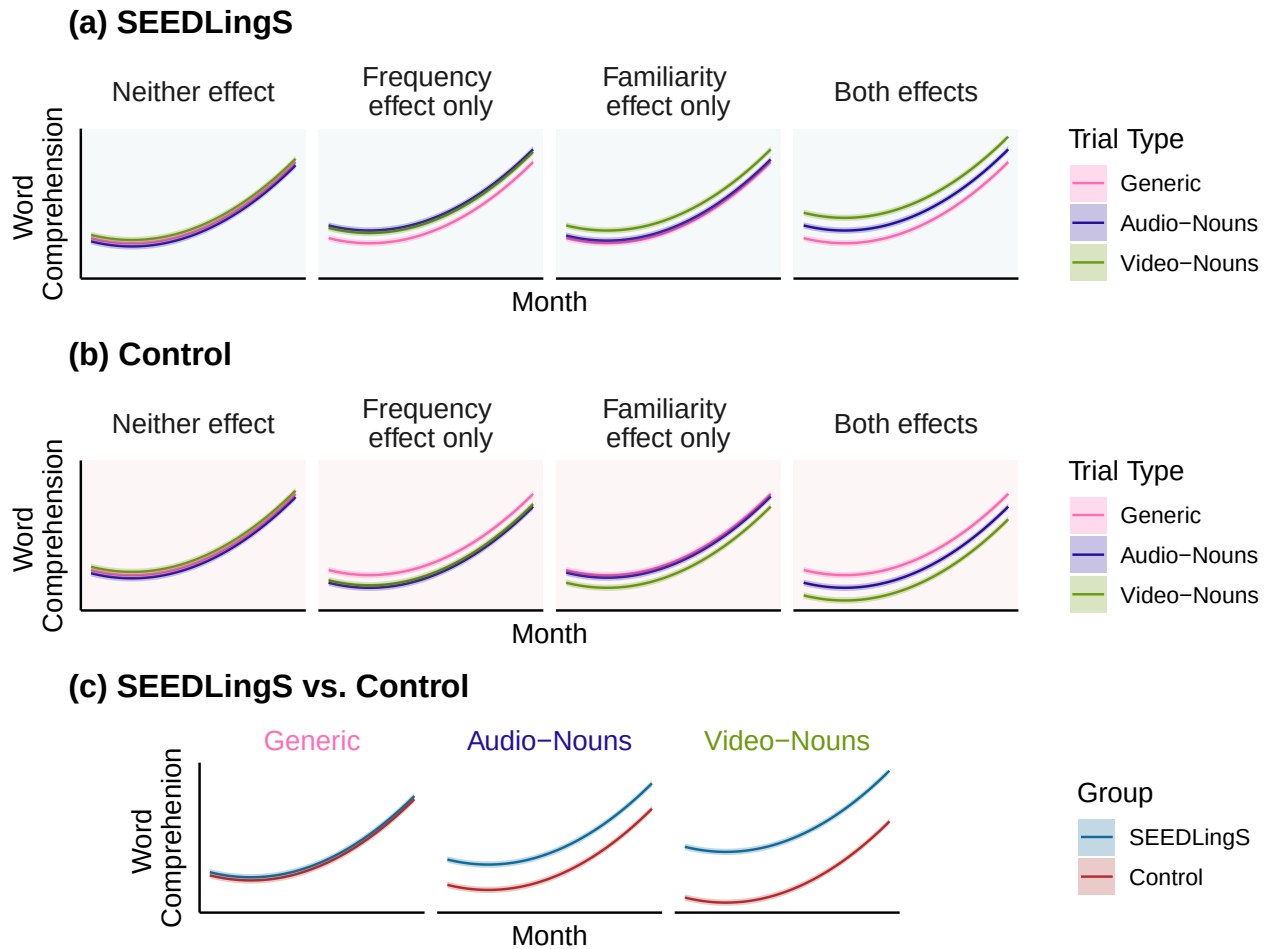
Here we investigate how experience shapes infants' early word comprehension and probe infants' early lexical representations. While these topics are not new, our approach to asking them is. Using the looking-while-listening eyetracking paradigm, we tested two groups of infants across two experiments on their understanding of three types of concrete, imageable nouns that varied in word frequency and referent familiarity. See Fig. 1 for study design schematic.



recordings and in-lab eyetracking every two months (SEEDLingS, detailed below). SEEDLingS infants were tested on two categories of word-image pairs for each eyetracking study: (1) common concrete nouns used for all infants, depicted with common baby-friendly images of their referents (“*generic*” items; 4 yoked pairs of 8 items), (2) concrete nouns hand-picked for each infant based on their high frequency in a given child’s previous two months’ home recordings (“*hand-picked*” items). The hand-picked items were further subdivided into two types: a set picked from the previous two months’ audio recordings, depicted with common baby-friendly images of their referents (“*audio-nouns*”; 2 yoked pairs of 4 items), and a set from the previous two months’ hourlong video recordings, depicted with images of the referents as they appeared in the videos (e.g., a given child’s specific water bottle; “*video-nouns*”; 2 yoked pairs of 4 items). Thus, each SEEDLingS infant had a hand-tailored experiment at 6 lab visits from 8-18 months (see Fig. 1).

Based on prior work, we had several predictions for Study 1 (Fig. 2a). First, we expected an improvement with age, and more specifically, a nonlinear shift around 12-14 months, in keeping with prior work (Bergelson, 2020). Second, if word frequency based on recent months’ linguistic input predicts comprehension, we would expect better performance on the hand-picked, high-frequency audio- and video-nouns than on the generic items. However, if the frequency of relatively common concrete nouns does not vary too widely across households (or across the high frequency range we are operating within), we may observe similar performance across conditions. Third, in terms of referent familiarity (and setting aside frequency effects), a narrow-scope representation would predict an advantage for video-nouns (featuring familiar referents) over audio-nouns and generic items (featuring unfamiliar referents).¹ In contrast, a broad-scope representation

¹ We use the terms ‘familiar’ and ‘unfamiliar/novel’ to describe contrasts in infants’ relative experience with referents. ‘Familiar’ referents were observed in infants’ homes and thus likely seen more often. ‘Unfamiliar’ or ‘novel’ ones were not specifically selected from their home video recordings and thus less familiar. That said, infants may still have seen these items elsewhere, e.g., in other people’s homes.

**Figure 2**

Possible result patterns, including predicted trial-type comparisons for SEEDLingS participants (a), for control participants (b), and predicted comparisons between SEEDLingS and control participants (c). The x-axis denotes infants' age between 8-18 months, and the y-axis denotes infants' word comprehension performance in eyetracking tasks. The curves represent hypothetical growth in comprehension as a function of age and may differ by trial type (a, b) or participant group (c). See the main text for a full description of the predicted patterns.

would predict similar comprehension between audio- and video-nouns. If both frequency and familiarity effects are present, SEEDLingS infants should perform best on video-nouns, then audio-nouns, and finally generic items. If neither effect is found, performance should be similar across trial types.

The second group of infants (Study 2) served as a cross-sectional control for the SEEDLingS infants: For each hand-tailored SEEDLingS experiment, a control infant was tested at the same age as the corresponding SEEDLingS infant. This control group fulfilled two purposes. First, it let us evaluate whether some age-related improvements seen in the longitudinal SEEDLingS participants could be attributed to familiarity with the testing experience, lab, method, etc., since the 4 pairs of generic items were tested across all participants of a given age. Second, it let us assess the degree to which our hand-picked items were known and recognizable by other age-matched infants.

We predicted the following result patterns for control infants (Fig. 2b). First, as in Study 1, we expected nonlinear improvement with age. Second, we considered two possibilities regarding word frequency. If the hand-picked audio- and video-nouns were truly unique to the SEEDLingS households, they would likely be less frequent for control infants than the generic items, leading to poorer performance on audio- and video-nouns. Alternatively, if the hand-picked words were common across households, we would expect similar performance on generic and hand-picked audio-nouns. Third, all trial types involved unfamiliar referents for control infants. However, while generic items and audio-nouns were paired with images of unfamiliar, prototypical referents, video-nouns featured idiosyncratic referents from other infants' homes, which may be more difficult to recognize. We therefore predicted lower performance on video-noun trials for control infants than generic trials. If control infants were influenced by both word frequency and referent properties, they should perform best on generic items, followed by audio-nouns, and then video-nouns. If neither effect was present, performance should be similar across trial types.

Finally, we compare SEEDLingS and control infants across trial types (Fig. 2c). For generic items, we expect similar performance across groups, unless familiarity with the experimental procedure boosts performance in the longitudinal SEEDLingS participants. For audio-nouns, which were selected to be high-frequency specifically for SEEDLingS infants, we expect better performance in that group. For video-noun trials, which contained both more frequent words and more familiar referents for SEEDLingS infants, we expect the largest group difference, with SEEDLingS infants showing a clear advantage in comprehension.

1.3.1 Data Availability

See osf.io/m2kdz for all stimuli, data, and code.

2 Study 1: SEEDLingS (Longitudinal)

2.1 Methods

2.1.1 Participants

Infants were recruited from a database in Rochester, NY at the University of Rochester. Families received \$340 and a small gift for participating in the entire yearlong study of word learning (the Study of Environmental Effects of Developing LINGuistic Skills, i.e., SEEDLingS). Beginning at 6 months, infants were audio recorded using LENA for a full day and video recorded for an hour each month, on separate days (see Bergelson, 2017; Bergelson et al., 2018 for data and description). Data collection took place from 2014-2016.

Based on prior work, we targeted a sample size of 48, given components of the broader study that split the group into three experimental conditions of $n = 16$ each (the standard minimum sample size in the field). We enrolled and retained 44 infants over an 8-month enrollment window.² Therefore, the final SEEDLingS sample consisted of 44 infants (21 girls), including one set of dizygotic twins. All infants were healthy, and parents

² Two more families enrolled but opted out after the first set of visits; they are excluded here.

reported no hearing or vision concerns. All were carried full term (40 ± 3 weeks), and heard at least 75% English at home. All families were consented in a process approved by University IRB. The sample was 95% Caucasian and 5% multi-racial. 50% of the infants' mothers held a graduate degree, 41% held a bachelor's or an associate's degree, 7% completed some college, and 2% held a high school degree.

The data presented here come from the primary eyetracking study that these infants participated in every two months from 8-18 months. Among the 264 visits (6 per participant), eyetracking data from 10 visits were excluded due to fussiness, computer issues, experimenter error, or parental interference, which resulted in usable data for < 25% of trials (see trial exclusion criteria in the Results section). There were also secondary experiments that are beyond the scope of the present work; these always followed the initial experiment reported here.

2.1.2 *Materials*

2.1.2.1 *Eyetracking Stimuli.* At each of the six visits, SEEDLingS participants were tested on 16 words, split into three types: generic nouns (8 words), audio-nouns (4 words), and video-nouns (4 words), organized into 4, 2, and 2 yoked pairs, respectively. See Fig. 1 for a schematic of the design and sample stimuli. The full set of stimuli is available at osf.io/m2kdz.³ Additionally, during a set of warm-up trials, each participant saw four additional warm-up items (one at a time) that did not overlap with the test items (e.g., apple, car, juice, spoon, bird).

Generic nouns were selected from a predetermined list based on past infant word comprehension studies. These were paired with baby-friendly, prototypical images from a

³ We quantified the amount of test item overlap across infants and within infants across months at both the noun and test-pair levels. Although some nouns were reused, exact test-pair reuse was negligible (<10% across infants, <1% within infants across months); full summary tables are provided in a supplemental file on OSF for curious readers at osf.io/m2kdz/files/4qcqh.

lab database that is part of the SEEDLingS Corpus (Bergelson, 2017). The same set of generic items was shown to all infants at a given timepoint, i.e., the images were not selected based on frequency or familiarity to specific infants.

In contrast, audio-nouns and video-nouns were hand-picked for each infant based on their individual home recordings and were thus unique to each participant. Audio-nouns were high-frequency concrete nouns annotated in each infant’s audio recordings from the previous two months. For 6-7 months, the full day was annotated to extract these items. For subsequent months, we algorithmically determined the ‘highest talk’ hours (top 4 for 8-13 months, and top 3 for 14-17 months), which were then annotated for concrete nouns. These annotations were then used to select high-frequency nouns for testing (two from each of the preceding two months’ audio-recordings), yielding 4 target words per visit. These nouns were paired with baby-friendly, prototypical, high-resolution images found online or already in the lab database, similar to the images used for generic items. For example, for SEEDLingS infant X’s 14-month visit, we selected ‘drum’ and ‘shoe’ from audio recordings at 12 months and ‘ball’ and ‘hand’ from 13 months, then paired each with a corresponding image that was unfamiliar to infant X.

Video-nouns were also high-frequency concrete nouns, but drawn from each infant’s hour-long video recordings in the previous two months. All concrete nouns were annotated, and we selected two high-frequency nouns from each of the preceding two months’ videos, yielding 4 target words per visit. Each image selected as the referent for a video-noun was selected based on the infant’s video recordings. That is, we obtained a high-resolution image of the same referent (either online or supplied by the parent if the item was unique to the home, e.g., a family pet). For instance, for one infant at 14 months, we selected ‘egg’ and ‘cow’ from the video at 12 months and ‘water’ and ‘baby’ from the video at 13 months. ‘Egg’ was paired a photo of fried eggs as seen on the infant’s high chair tray, ‘water’ with their pink water bottle, and ‘cow’ and ‘baby’ with cartoon characters that the child had

seen in a book within each video.

This approach prioritized fidelity to infants’ actual recent experience: because video-noun referents were selected based on infants’ home video recordings from the two preceding months, we had direct evidence that these were referents the infant had recently encountered and were familiar with. At the same time, this design meant that video-noun referents were sometimes more idiosyncratic, and potentially less prototypical, than the images used for generic and audio-noun trials. For example, an image of scrambled eggs may seem less prototypical to adults than an egg in its shell, but may depict the most common way at least some infants encounter eggs. That is, we aimed to select test images that depict the intended referents in a way we expected to be readily recognizable for a given infant.⁴

Each target word was embedded in a sentence recorded by a female speaker using infant-directed speech. The sentences, ranging from 0.97 to 2.80 s in length, were normalized to 72 dB using Praat. Each sentence used one of four carrier phrases: “Where is the X?”, “Look at the X!”, “Do you see the X?”, and “Can you find the X?” (where X represents the target word). The same carrier phrase was used within each item-pair. During the warm-up trials, sentences were played over computer speakers. During test trials, only the parent heard the pre-recorded sentences, which they then repeated aloud to

⁴ When constructing the test pairs, we generally aimed to avoid saliency mismatches. Where possible, we also tried to avoid pairing items from the same superordinate semantic category, which could also disrupt word recognition. This was relatively straightforward for the preplanned generic trials. It was more difficult for the audio- and video-noun trials because these hand-picked items were selected separately for each infant and testing session from a limited set of high-frequency candidate words (after excluding the predetermined generic items), while also satisfying other constraints such as avoiding saliency mismatch. To assess the extent of this issue, we manually coded superordinate semantic relatedness for all generic pairs and for a random 10% subsample of hand-picked pairs. None of the generic pairs involved items from the same superordinate category, whereas around 15% of audio-noun pairs and 15% of video-noun pairs did.

their infant (see Procedure for details). All images were edited onto a plain gray background.

Computing Test Word Frequency. 2.1.2.1.1 We computed frequency using the following procedure. For each test word, we computed *relative* (rather than absolute) frequencies to control for variability in the total noun input across infants. Frequencies were computed at the individual level, based on what each infant heard in the two months prior to each visit, rather than pooled across all infants, since different nouns were queried for different infants. For each infant, the frequency for each queried noun was calculated as its proportion out of all noun tokens spoken in this infant’s annotated recordings from the two months preceding the visit (total number of noun tokens in these annotated samples: $M = 1064$, $SD = 494$, range: 191-3053).⁵ Specifically, to account for the varying amount of annotated data across months, we based our frequency estimation on the top 3 (algorithmically determined) “high talk” hours of audio and the hour-long video recordings, excluding infants’ own production. Instances of these nouns were excluded if they occurred outside of the top 3 hours of audio recordings, in other months, or in unannotated segments.⁶ Because our operational measure of word comprehension is defined at the test pair level (detailed in Section 2.2.1), we computed test pair frequencies by averaging the

⁵ In the Supplementary Materials, we examined alternative operationalizations of frequency that varied both the temporal window over which frequency was computed (two preceding months, all preceding months, or all months) and the level of aggregation across infants (within infant vs. pooled across infants). Results showed that frequency measures computed within infants were highly consistent across temporal windows and yielded similar patterns in predicting word comprehension, whereas frequency measures pooled across infants were less strongly aligned with the original measure and were less predictive of performance.

⁶ In the Supplementary Materials, we compared frequency estimates for test words at 8 months based on the top 3 highest-talk audio hours with estimates derived from full daylong audio recordings. Results showed that the highest-talk segments closely approximate the frequency distributions obtained from larger samples and preserve the frequency contrast between hand-picked and generic test nouns, indicating that this sampling procedure provides a reliable estimate of infants’ noun input.

individual frequencies of the two words comprising each test pair.

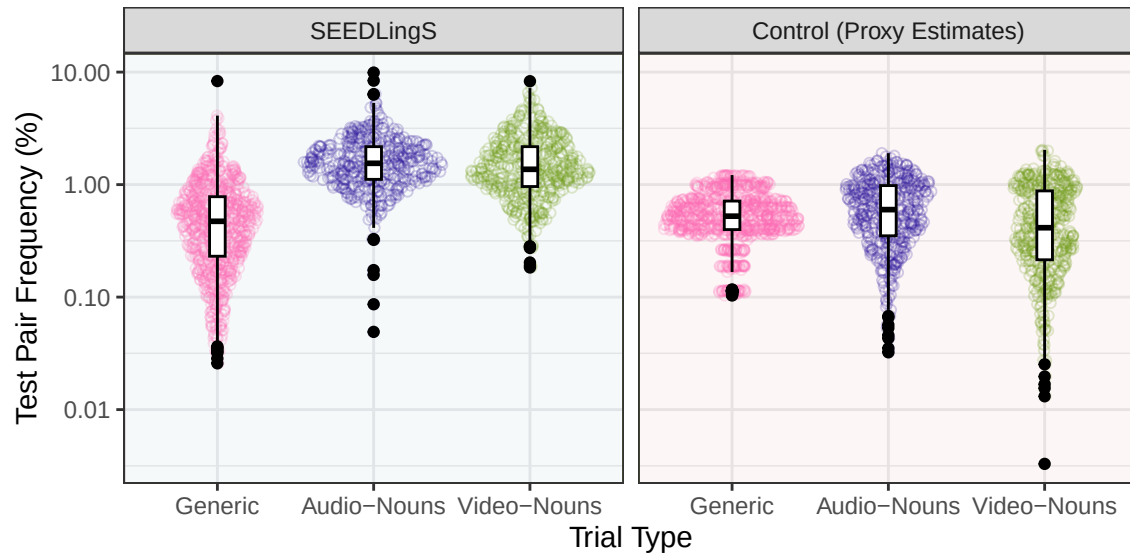


Figure 3

Distribution of frequency by trial type (generic, audio-nouns, video-nouns) for SEEDLingS (left) and control participants (right). Each empty dot represents one test pair for a participant at a given visit. For SEEDLingS participants, word frequency was computed as the proportion of a given noun out of all noun tokens annotated in that infant’s home recordings from the two months preceding the visit. Audio- and video-nouns were more frequent than generic nouns. For control participants, word frequency was estimated using proxy values: the frequency of each noun across all SEEDLingS participants, excluding the one each control child was paired with. Frequency was similar across all three trial types for control infants.

To illustrate, consider the SEEDLingS infant in the example above who, at their 14-month visit, who was tested on the yoked pair ‘egg-water’. Across recordings from 12 and 13 months, ‘egg’ was said to them 27 times, and the total number of noun tokens heard was 1,429. The relative frequency for ‘egg’ was thus $(27 / 1,429) \times 100\% = 1.89\%$. Following the same procedure, the relative frequency of ‘water’ was $(68 / 1,429) \times 100\% = 4.76\%$. The test pair frequency was computed as the average of the two: $(1.89\% + 4.76\%) / 2 = 3.32\%$. As visualized in Fig. 3 (left panel) and detailed in the Supplementary Materials, test pairs composed of hand-picked audio- and video-nouns were similarly high

in frequency, and both were more frequent than the generic test pairs.

To summarize, we manipulated word frequency and referent familiarity across the three trial types. Generic trials featured a predetermined list of concrete nouns paired with novel, prototypical referent images. Audio-noun trials included high-frequency concrete nouns drawn from each infant’s audio recordings from the two months prior to each visit, paired with novel, prototypical referents. Video-nouns also involved high-frequency concrete nouns, but drawn from each infant’s video recordings over the same period, and were paired with familiar referents from each infant’s home environment. Therefore, by design, audio- and video-nouns were more frequent than generic nouns, and video-nouns were paired with more familiar referents than both audio- and generic nouns. To select the high-frequency audio- and video-nouns and the familiar referents for video-nouns, we focused on the two months preceding each experimental session so that the stimuli would reflect infants’ recent experience and be more likely to be recognizable to each infant. Supplementary analyses show that the frequency results are robust to alternative temporal windows (see the Supplementary Materials section “Frequency Estimates Across Temporal Windows and Aggregation Levels”). For familiarity, however, this choice should be understood primarily as a stimulus-selection decision rather than as a claim that referent familiarity operates on that specific timescale.

2.1.2.2 Parent Surveys. Parents completed two surveys monthly throughout the yearlong study (i.e., 6-18 months): (1) the CDI Words & Gestures Form, and (2) motor questionnaires (Moore et al., 2019), which included the gross motor scale from the Early Motor Questionnaire (EMQ, Libertus & Landa, 2013) and the brief locomotor questionnaire from Walle and Campos (2014). After the initial administration, subsequent surveys were pre-filled with the responses from the previous form. For the present study, we report only the six CDI administrations that aligned with the eyetracking timepoints (every two months, 8-18 months; see Section 3.2.3.2). Results of the motor questionnaires are reported in Moore et al. (2019).

At each visit, parents also completed the Word Exposure Survey, which asked parents to rate how often they believed their child heard the 16 words tested in the experiment on a five-point scale (1 = “Never”, 5 = “Several times a day”). See Dong et al. (2026) for more details.

2.1.3 Procedure

We assessed infants’ word comprehension using an adapted looking-while-listening paradigm, following a similar procedure in prior work (Bergelson & Aslin, 2017a; Bergelson & Swingley, 2012). After completing paperwork, the infant and parent were escorted to a dimly lit testing room. The infant sat on their parent’s lap in front of a computer screen. To collect infants’ visual fixation data, we used an Eyelink 1000+ Eyetracker camera (SR Research) mounted on a movable arm at the bottom of the screen. The eyetracker was sampling monocularly at 500 Hz, with an average accuracy of 0.5°. A small, high-contrast sticker on the infant’s forehead helped the eyetracker keep the infant’s position.

Following calibration, the infant saw four warm-up trials, each of which displayed a single image on the screen accompanied by a pre-recorded sentence. After warm-up, each infant completed 32 test trials. Half of the trials included generic items, and the other half contained hand-picked items evenly split between audio-nouns and video-nouns. Each of the 16 test items appeared twice as the target word and twice as the distractor word.

During each test trial, the infant viewed two images on a gray background, one on each side of the screen. The parent wore a visor that blocked their view of the screen, and headphones, over which they heard the test sentence and a beep that indicated they should repeat the sentence aloud for the child. The target word occurred 3.78 s on average after the trial began ($SD = 0.88$). An experimenter pressed a button to mark when the parent said the target word, after which the images remained on screen for another 5 s (see Fig. 4).

The experimenter had the option to recycle a test trial either immediately or after

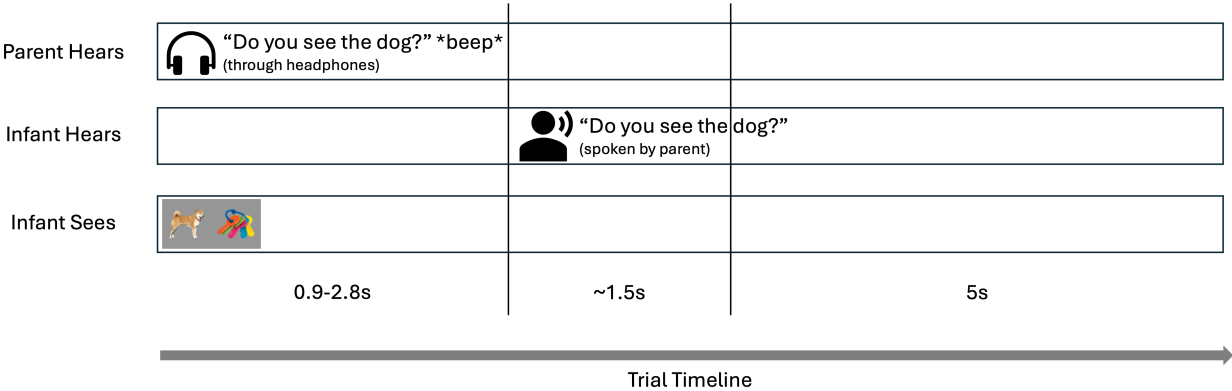


Figure 4

Sequence of one test trial. During each test trial, parents listened to a test sentence through headphones, followed by a beep. After hearing the beep, they repeated the sentence aloud to their infant. Images were shown to infants starting from the beginning of the trial and remained visible until 5 s after the parent began saying the target word.

completing all planned trials if the initial attempts were deemed likely unusable due to fussiness. For each planned trial with multiple attempts, we retained the most recent usable attempt. See the Supplementary Materials for details.

Trial order was pseudo-randomized to present consecutive repetitions of the same test pair or carrier phrase, and to ensure that target images did not appear on the same side of the screen more than twice in a row. The left-right placement of target images was counterbalanced within test pairs across trials. To sustain infants' interest, a 2-s animation featuring colorful shapes, smiley faces, and bird-chirping sounds was played as needed.

2.1.4 Data Analysis

We used R [Version 4.2.1; R Core Team (2022)] for all our analyses. The full list of R packages and versions can be found in the Supplementary Materials.

We adopted a mixed-effects modeling approach that begins with a full model motivated by our hypotheses. Models were fit using the `lme4` R package (Bates et al., 2015) and *p*-values were obtained using the `lmerTest` package (Kuznetsova et al., 2017).

For models involving trial-level eyetracking data, we included random intercepts for subject and test pair. We then used `lmerTest::step()` (Kuznetsova et al., 2017) to perform backward stepwise model selection on both fixed and random effects, seeking the simplest model that adequately captures the data. In all cases, we report the best-fitting model after model selection.

Categorical predictor variables were coded using simple (effect) coding. Like dummy coding, this approach compares each level of a predictor to a reference level; however, unlike dummy coding, the intercept in effect-coded models represents the grand mean rather than the mean of the reference group.

2.2 Results

We present two sets of analyses below. First, to examine the influence of age, word frequency, and referent familiarity, we compared SEEDLingS participants' word comprehension across trial types and age groups. Word frequency and referent familiarity were manipulated between trial types: Audio-noun and video-noun trials featured higher-frequency words than generic trials, and only video-noun trials paired target words with familiar referents drawn from infants' home environments. To foreshadow our results, this initial analysis revealed a moderate effect of frequency (audio-nouns and video-nouns > generic) but no reliable effect of familiarity. To build on these findings, our second analysis goes beyond trial-type comparisons by collapsing across trial types and using continuous frequency estimates based on infants' home recordings to directly predict eyetracking outcomes.

2.2.1 *Eyetracking Data Preparation and Exclusion*

We grouped the raw fixation data into 20-ms bins, with time 0 marking the onset of the target word. The post-onset analysis window was set from 360 to 3,500 ms. The standard 367-ms starting time provides sufficient time to initiate eye movements in response to the target word (Fernald et al., 1998); we rounded this to 360 ms to align with

the 20-ms bins. The extended 3,500-ms offset, instead of the typical 2,000-ms offset, accommodates the slower reaction times of the younger age groups (Bergelson & Swingley, 2012). Following Bergelson and Swingley (2012), we operationalized word comprehension performance as the difference of fixation proportions (referred to as the *increase in target looking* hereafter): For each yoked pair A-B, we calculated the mean proportion of fixation on picture A when it served as the target (with B as the distractor), then subtracted the mean proportion of fixation at picture A when it acted as the distractor (with B is the target). For example, for the test pair ‘banana-kitty’, a child’s performance was assessed by comparing how much more she looked at the banana image when ‘banana’ was named as the target versus when ‘kitty’ was the named target. A difference score of 0 indicates chance performance, i.e., the child looked equally at the two test images regardless of the named target. Positive difference scores indicates greater looking at an image when it is the named target than when it is the distractor, taken to indicate word comprehension.

Before presenting the detailed numbers involved in the exclusion process, we first provide an overview of our approach. To arrive at the final dataset, we first removed trials in which participants did not hear the planned stimuli due to infant fussiness/crying, experimenter error, parent’s failure to say the target word, or technical difficulties. Next, we applied a data-driven approach to exclude trials with insufficient fixation data, specifically those in which participants failed to fixate on either the target or distractor for more than 1/3 of the analysis window. For planned trials with multiple attempts remaining after exclusion, we retained the most recent usable attempt. Finally, we excluded participants who contributed data to fewer than 25% of planned trials, along with any remaining trials from these participants.

Our final SEEDLingS dataset included 1,698 test pairs, which occurred within 6,165 trials across 254 infant visits ($M = 24.27/32$ trials, $SD = 6.94$, range: 8-32). We originally planned for 8,448 trials in total ($44 \text{ subjects} \times 6 \text{ visits} \times 32 \text{ planned trials per visit}$),

corresponding to 2,112 test pairs (each pair appeared on 4 trials, with each item as the target twice). We lost 27% of test trials but only 20% of test pairs, since the loss of a single trial did not necessarily prevent computation of pair-level eyetracking outcomes.

Among the 8,448 planned trials, 60 were skipped as the experiment ended early due to infant fussiness/crying. We then removed 2,182 trials in which participants did not hear the planned stimuli or failed to provide sufficient fixation data. Finally, we excluded 10 infant visits that contributed too few data ($< 25\%$ of planned trials), along with an additional 41 remaining trials from these visits. See the Supplementary Materials for additional details on trial and participant exclusion criteria, including counts of repeated attempts and specific reasons for exclusion.

2.2.2 *SEEDLingS Participants' Word Comprehension by Trial Type and Age*

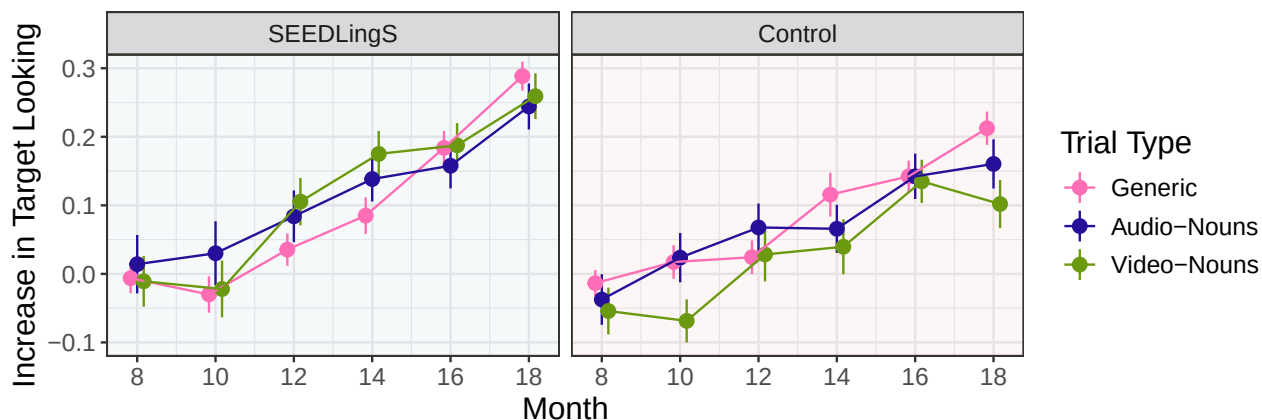


Figure 5

Word comprehension among SEEDLingS (left) and control participants (right) by trial type (generic, audio-nouns, video-nouns). The Y-axis represents the increase in target looking, i.e., the increase in fixation when an image is the target within a yoked pair compared to when it is the distractor within the same pair. Dots show means, vertical lines show standard errors. SEEDLingS participants showed increased word comprehension over time, with the largest gap between generic trials and the hand-picked trials (audio- and video-nouns) at ~12-14 months. Control participants also improved in word comprehension over time, though performance on video-nouns consistently lagged behind the other trial types.

To examine whether word frequency and referent familiarity (manipulated between trial types) influence infants' word comprehension, and whether these effects interact with age, we fit a linear mixed-effects model. The model predicted the increase in target looking as a function of trial type and its interaction with age. Based on prior findings that infants' online word comprehension improves nonlinearly with age starting at around 12 months (Bergelson, 2020), we included both linear and quadratic age terms. Age was centered at 12 months and scaled such that each unit reflects a 2-month change in chronological age. Trial type was simple-coded with generic trials as the reference level.

The full model included fixed effects of scaled age (linear and quadratic), trial type (generic, audio-nouns, video-nouns), and their interactions. Model selection retained all fixed effects except for the interaction between trial type and linear age (final model: `increase in target looking ~ age-scaled + trial type × age-scaled2 + (1|subject)`). Model residuals were normal by a Shapiro-Wilk test ($W = 1, p = 0.11$).

The final model revealed significant main effects of both linear age and quadratic age, consistent with a nonlinear developmental trajectory in word comprehension, at least for the generic test items. For example, predicted values derived from the fixed effects suggested that comprehension of generic words (operationalized as increase in target looking) increased by approximately 0.01 unit between 8-10 months, and 0.06 unit between 12-14 months.

Trial type also significantly predicted word comprehension: Both audio-noun and video-noun trials led to greater increase in target looking compared to generic trials, suggesting a frequency-based advantage. However, there was no evidence that video-nouns led to significantly better comprehension than audio-nouns, indicating no clear benefit of referent familiarity. In fact, the coefficients for video-nouns and audio-nouns (both contrasted against generic trials) were similar in magnitude and standard error, suggesting that the familiarity manipulation did not yield a measurable advantage above the

Table 1

Fixed effects from the best-fitting model of SEEDLingS participants' word comprehension as a function of trial type and age in months.

Term	Estimate	SE	t	df	p-value
(Intercept)	0.048	0.017	2.72	65.22	0.008
Month-Scaled	0.049	0.005	9.61	1,660.19	<.001
Month-Scaled ²	0.008	0.003	2.65	1,657.31	0.008
Trial Type (Audio-Nouns)	0.051	0.025	2.00	1,646.95	0.046
Trial Type (Video-Nouns)	0.060	0.025	2.46	1,645.22	0.014
Trial Type (Audio-Nouns) $\times$ Month-Scaled ²	-0.011	0.005	-2.03	1,645.00	0.042
Trial Type (Video-Nouns) $\times$ Month-Scaled ²	-0.010	0.005	-1.98	1,644.19	0.048

Note. Trial type is simple coded with generic trials as the reference level. Random effects are (1|Subject). See text for model selection details.

frequency effect.

Additionally, the significant interactions between quadratic age and trial type indicate that the frequency-based advantage (i.e., the difference between audio- or video-nouns and generic trials) varied with age. The negative coefficients for these interactions suggest that the frequency-based advantage was strongest in the middle of the 8-18-month age range (i.e., around 12-14 months) and weaker at the younger (8-10 months) and older (16-18 months) ends of the age range. For example, according to the model's fixed effects, audio-nouns led to a 0.05-unit increase in word comprehension compared to generic items at 12 months, but this frequency-based advantage shrunk to only 0.01 unit at 8 and 16 months. See Table 1 for the full model outputs.

Together, these results suggest that word frequency, but not referent familiarity,

facilitated SEEDLingS infants' word comprehension. Moreover, the frequency effects appear strongest around the first birthday. This is consistent with Fig. 5a, which shows the largest gap in comprehension between generic items and the higher-frequency, hand-picked items (audio-nouns and video-nouns) at 12-14 months. However, the observed frequency effects were modest in size (with p -values between .04 and .05), prompting a follow-up analysis that takes a finer-grained approach: Rather than comparing broad trial types, we examine how continuous, corpus-based frequency estimates predict comprehension across all items.

2.2.3 *Finer-Grained Analyses on Frequency Effects*

As described in the Methods section, we estimated word frequency for each SEEDLingS infant as the proportion of each test word relative to all noun tokens in that child's home recordings from the prior two months. Because our word comprehension outcome is defined at the test pair level (i.e., increase in target looking), we calculated test pair frequency as the average frequency of the two words in each pair. We log-transformed this value for analysis due to the skewness of untransformed frequency data, which required us to exclude data from 55 test pairs (out of 1,698) with zero frequency.

We first fit a linear mixed-effects model predicting increase in target looking from log-transformed test pair frequency, linear and quadratic age, and their interactions. Model selection retained all fixed effects (final model: `increase in target looking ~ log test pair frequency × (age-scaled + age-scaled2) + (1 | subject)`). Model residuals were normal by a Shapiro-Wilk test ($W = 1$, $p = 0.15$).

The final model revealed that higher test pair frequency significantly predicted greater increase in target looking, suggesting that infants are more likely to understand words they have heard more frequently in their home environments. Both linear and quadratic age terms were significant, consistent with the nonlinear developmental trajectory observed in the previous analysis. Importantly, we also found significant

Table 2

Fixed effects from the best-fitting model of SEEDLingS participants' word comprehension as a function of log test pair frequency and age in months.

Term	Estimate	SE	t	df	p-value
(Intercept)	0.051	0.018	2.91	60.56	0.005
Log Test Pair Frequency	0.109	0.025	4.43	1,601.40	<.001
Month-Scaled	0.050	0.005	9.47	1,606.25	<.001
Month-Scaled ²	0.007	0.003	2.44	1,604.59	0.015
Log Test Pair Frequency $\times$ Month-Scaled	0.027	0.012	2.38	1,599.10	0.018
Log Test Pair Frequency $\times$ Month-Scaled ²	-0.025	0.006	-3.88	1,600.83	<.001

Note. Random effects are (1|Subject). See text for model selection details.

interactions between frequency and age terms (positive linear age $\times$ frequency interaction, negative quadratic age $\times$ frequency interaction), indicating that frequency effects were amplified at intermediate ages (i.e., between 12-14 months.) For example, according to the model's fixed effects, at 12 and 14 months, every one-unit increase in log-frequency corresponded to a 0.11-unit gain in word comprehension (operationalized as increase in target looking). By contrast, at 10 and 16 months, the effect of log-frequency was smaller, with each unit increase in log-frequency yielding only a 0.06-unit gain in word comprehension. See Table 2 for the full model outputs.

To further probe these effects, we fit separate models at each age to test whether log test pair frequency predicted comprehension within each month. These by-month analyses showed significant positive effects of frequency at 12 and 14 months ($ps < .007$), but not at earlier or later ages ($ps > .07$; Fig. 6).⁷ Indeed, log frequency is more positively correlated

⁷ Results at 12 and 14 months remain significant even with conservative Bonferroni correction for six comparisons ($\alpha = .0083$).

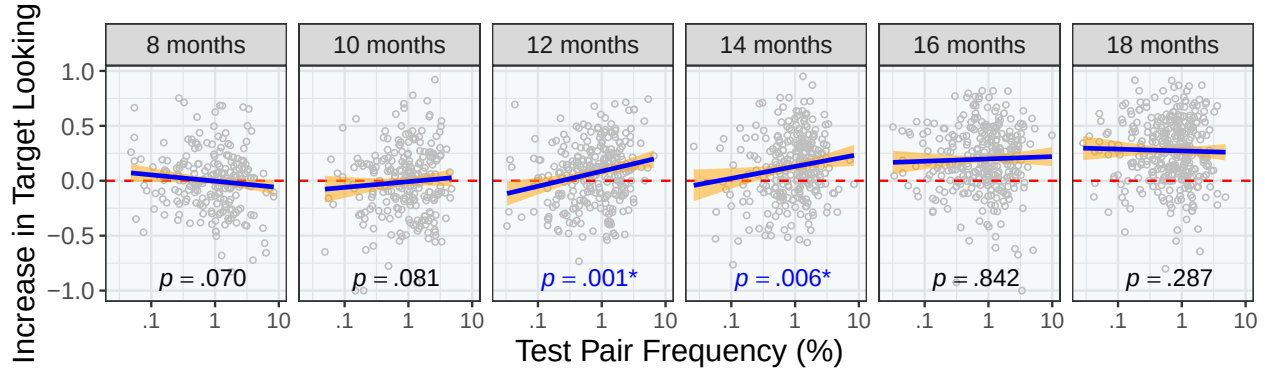


Figure 6

Nonlinear shift in the relationship between word comprehension (y-axis) and frequency (x-axis) over 8-18 months from the best fitting model: $\text{comprehension} \sim \log\text{freq} \times (\text{age} + \text{age}^2) + (1/\text{subj})$. Each dot corresponds to a single test pair from one infant. Frequency significantly predicted comprehension only at 12 and 14 months (cf. p-values in each panel). Robust linear regression lines are in blue, 95% CIs in orange. The red dashed line ($y = 0$) indicates chance-level performance.

with comprehension at 12 ($r = 0.20$, $p = 0.001$) and 14 months ($r = 0.16$, $p = 0.007$) than at the other months ($r_{\min} = -0.12$, $r_{\max} = 0.10$, $ps > 0.07$).

To clarify how the null effects at the younger and older age groups should be interpreted, we also tested whether overall performance at each age differed from chance. For each month, we fit a separate intercept-only model predicting increase in target looking and tested whether the intercept differed from 0, where 0 indicates chance performance. Results showed that infants were at chance at 8-10 months ($p > 0.86$), but were reliably above chance at 12-18 months ($p < 0.004$). We therefore interpret the absence of frequency effects at 8-10 months cautiously, as performance at these ages was at floor, whereas the null effects at 16-18 months arose after comprehension had become robust across items.

Taken together, these findings provide converging evidence that infants' word comprehension is influenced by word frequency, but not by referent familiarity in their

home environment. In the first analysis, infants understood the higher-frequency audio- and video-nouns better than the lower-frequency generic nouns, with effects peaking around the first birthday. However, comprehension did not differ between audio- and video-noun trials despite the referent familiarity manipulation, suggesting that infants' early noun knowledge extends beyond specific familiar referents. In the second analysis, we moved beyond trial-type comparisons and found that a continuous, corpus-based estimate of word frequency robustly predicted comprehension, particularly at 12 and 14 months. This finer-grained analysis confirmed and extended the frequency effect, suggesting that infant's early word comprehension benefits from more exposure to specific word forms. We return to the nonlinear shift in frequency effects over 8-18 months and explore potential mechanisms in the Discussion section.

3 Study 2: Cross-Sectional Control

In Study 1, we collected a longitudinal sample of 44 infants and hand-tailored eyetracking experiments based on their home recordings to assess how experience shapes early word comprehension. Each infant was tested at six timepoints (every two months, 8-18 months). We found that infants were better at comprehending higher-frequency words, particularly at intermediate ages (12-14 months). We also observed that infants extended nouns equally well whether they were depicted as familiar referents from their home environments or as prototypical (but not child-specific) referents.

However, two questions remained. First, to what extent would the items hand-picked for SEEDLingS participants be equally well understood by other infants? Second, could repeated in-lab testing have influenced the longitudinal infants' word comprehension performance? To address these questions, Study 2 tested a group of cross-sectional control infants who were age- and gender-matched to the longitudinal SEEDLingS participants at each visit. These control infants completed the same eyetracking tasks, which contained stimuli hand-picked for other infants (i.e., not tailored

to them; Fig. 1).

3.1 Methods

3.1.1 *Participants*

A separate group of 247 infants (118 girls) were recruited from a database in Durham, N.C. at Duke University. Families received \$5 or \$10 (depending on the distance traveled) and a small gift for participating. Each control participant was matched to a single participant from the SEEDLingS study, based solely on their age and gender. That is, for each individual eyetracking session for a SEEDLingS participant, a control participant of the same gender and within one week of the target age completed the same in-lab assessment.⁸ This resulted in 6 separate control participants for each SEEDLingS infant, one for each timepoint. An additional 17 infants were excluded due to fussiness, computer issues, experimenter error, or parental interference, which resulted in usable data for < 25% of trials (see trial exclusion criteria below).

The final sample was 79% Caucasian, 6% Black, 2% Asian, 11% multi-racial; another 2% were identified as “other”, and the remaining 3 infants’ parents did not provide racial information. 54% of the infants’ mothers held a graduate degree, 37% held a bachelor’s or an associate’s degree, 4% completed some college, 2% held a high school degree, and 1% completed some high school; the remaining 4 infants’ parents did not provide information on maternal education.

All infants were healthy, and parents reported no hearing or vision concerns. All were carried full term (40 ± 3 weeks), and heard at least 75% English at home. All families were consented in a process approved by University IRB.

⁸ Due to recruitment logistics error, 9 control participants were not gender-matched to their corresponding SEEDLingS participants.

3.1.2 *Materials and Procedure*

Each control participant completed the same eyetracking experiment, using the same set of stimuli and testing procedure as the SEEDLingS participant to whom they were matched. In addition, at each visit, parents completed the CDI (see Section 3.2.3.2) and the Word Exposure Survey (results reported in Dong et al., 2026). Note that CDI administration differed between groups: for longitudinal SEEDLingS participants, CDI forms after the initial administration were pre-filled with responses from the previous CDI, whereas cross-sectional control participants completed the CDI only once and without pre-filled responses.

While we did not collect home recordings from the control sample and were thus unable to directly compute the frequency of the test words based on their home recordings, we were able to compute a proxy for what this value might have been using the recordings from Study 1. We estimated test word frequency for control infants using recordings from all SEEDLingS infants from the two months preceding the target age, excluding the one that the control infant was paired with. For example, suppose control infant A was paired with SEEDLingS infants X at 14 months, and both were tested on the yoked pair ‘egg-water’. Across recordings from 12 and 13 months, ‘egg’ was said a total of 170 times to all SEEDLingS infants combined except X. The total number of noun tokens heard by those same infants was 44,765 noun tokens. The proxy estimate of the frequency of ‘egg’ for control infant A was thus $(170 / 44,765) \times 100\% = 0.38\%$. The proxy estimate of the frequency of ‘water’ for control infant A was computed in the same way: $(548 / 44,765) \times 100\% = 1.22\%$. The test pair frequency for control infant A was then defined as the average of the two word frequencies: $(0.38\% + 1.22\%) / 2 = 0.80\%$.

As shown in Fig. 3 (right panel) and detailed in the Supplementary Materials, the three trial types did not differ systematically in frequency across months for control infants based on these proxy estimates. That is, audio- and video-nouns (i.e., high-frequency

concrete nouns hand-picked based on SEEDLingS infants’ recordings) were not consistently less frequent than generic nouns for control infants, despite not being selected with them in mind.

Thus, control infants viewed test pairs that we estimated were, to them, similar in word frequency (based on their occurrence collectively in all the non-target SEEDLingS recent months’ recordings). For controls, none of the trials featured familiar referent images. Generic and audio-noun trials used prototypical referents from the lab database. In contrast, video-noun trials used unfamiliar referents drawn from other infants’ idiosyncratic home environments.

3.2 Results

We present two sets of analyses below. First, to assess whether the test items hand-picked for SEEDLingS participants were understood by other age-matched infants, we analyzed control participants’ word comprehension by trial type across age groups. Next, to evaluate the potential impact of repeated in-lab testing on SEEDLingS participants’ word comprehension performance, we compare the two groups across several dimensions, including the amount of retained data, vocabulary scores, and word comprehension performance by trial type. For all statistical models involving group comparisons below, the variable “participant group” was simple coded with the control group as the reference level.

3.2.1 *Eyetracking Data Preparation and Exclusion*

Our final control dataset included 1,712 test pairs, which occurred within 6,151 trials across 247 infant visits ($M = 24.90/32$ trials, $SD = 6.22$, range: 8-32). We originally planned for 8,448 trials in total (44 SEEDLingS subjects $\times 6$ visits $\times 32$ planned trials per visit), corresponding to 2,112 test pairs (each pair appeared on 4 trials, with each item as target twice). We lost 27% of test trials but only 19% of test pairs, as the loss of one trial did not necessarily eliminate the corresponding pair-level measure.

Data from two visits were completely excluded due to parental interference or

inaudible parental speech. After excluding data from the two unusable visits (64 trials), another 178 trials were skipped as the experiment ended early due to infant fussiness/crying or technical issues. We then removed 2,002 trials in which participants did not hear the planned stimuli or failed to provide sufficient fixation data. Finally, we excluded 15 participants, along with an additional 53 remaining trials from these participants, as they contributed data for fewer than 8 out of 32 planned trials ($< 25\%$). We provide more details on trial and participant exclusion criteria, including counts of repeated attempts and specific reasons for exclusion, in the Supplementary Materials. A finer-grained breakdown of the number of retained trials across participant and age groups is provided in Section 3.2.3.1.

3.2.2 *Control Participants' Word Comprehension by Trial Type*

To examine control infants' ability to recognize words and referents drawn from others' experiences over time, we fit a linear mixed-effects model predicting the increase in target looking from trial type and its interaction with age terms. Similar to the model for trial-type comparisons among SEEDLingS infants in Section 2.2.2, the full model included fixed effects of scaled age (linear and quadratic), trial type (generic, audio-nouns, video-nouns), and their interactions.

Model selection retained trial type and linear age and dropped other fixed effects including quadratic age and the interaction terms (final model: `increase in target looking ~ trial type + age-scaled + (1|subject) + (1|test pair)`). A Shapiro-Wilk test on the residuals indicated a statistically significant deviation from normality ($W = 0.99$, $p < .001$); however, the test statistic is very close to 1, suggesting only a slight departure from normality. Given the large sample size ($n = 1,712$ test pairs), even minor deviations were likely to result in statistically significant p -values.

The final model revealed a significant main effect of linear age, suggesting that control infants improved in word comprehension with age. Specifically, for every two-month

Table 3

Fixed effects from the best-fitting model of control participants' word comprehension as a function of trial type and age.

Term	Estimate	SE	t	df	p-value
(Intercept)	0.042	0.011	3.95	152.26	<.001
Month-Scaled	0.041	0.006	7.02	219.74	<.001
Trial Type (Audio-Nouns)	-0.014	0.021	-0.68	66.86	0.5
Trial Type (Video-Nouns)	-0.054	0.021	-2.53	66.70	0.014

Note. Trial type is simple coded with generic trials as the reference level. Random effects are (1|Subject) + (1|Test Pair). See text for model selection details.

increase in age, control infants' word comprehension rose by 0.04 unit. However, contrary to Study 1, we found no evidence of nonlinear growth in word comprehension over time among control infants. We also found a significant contrast between video-noun trials and the other trial types (video-nouns < others), but not between audio-noun and the other trials, suggesting that control infants were overall worse at recognizing referent images for high-frequency nouns drawn from other infants' experiences. This is consistent with Fig. 5b, where performance on video-nouns lagged behind the other two trial types. See Table 3 for the full model outputs.

Notably, a similar pattern was observed in an adult control study in which adults viewed the same image pairs used in the infant experiments and selected the labeled referent (see the Supplementary Materials). Adults took significantly longer to identify the correct referents in video-noun trials compared to audio-noun or generic trials. This parallel between the adult and infant results suggests that control infants' lower performance on video-noun trials may reflect, at least in part, greater processing difficulty associated with

these child-specific referents. We return to this possibility in the Discussion.

3.2.3 Comparing *SEEDLingS* and Control participants

The goal of the following analyses is to compare word comprehension between the two participant groups across different trial types, in order to further evaluate potential effects of word frequency and referent familiarity. Before turning to these comparisons, however, we first assess whether the groups differed along other dimensions that could influence word comprehension performance but were unrelated to frequency or familiarity.

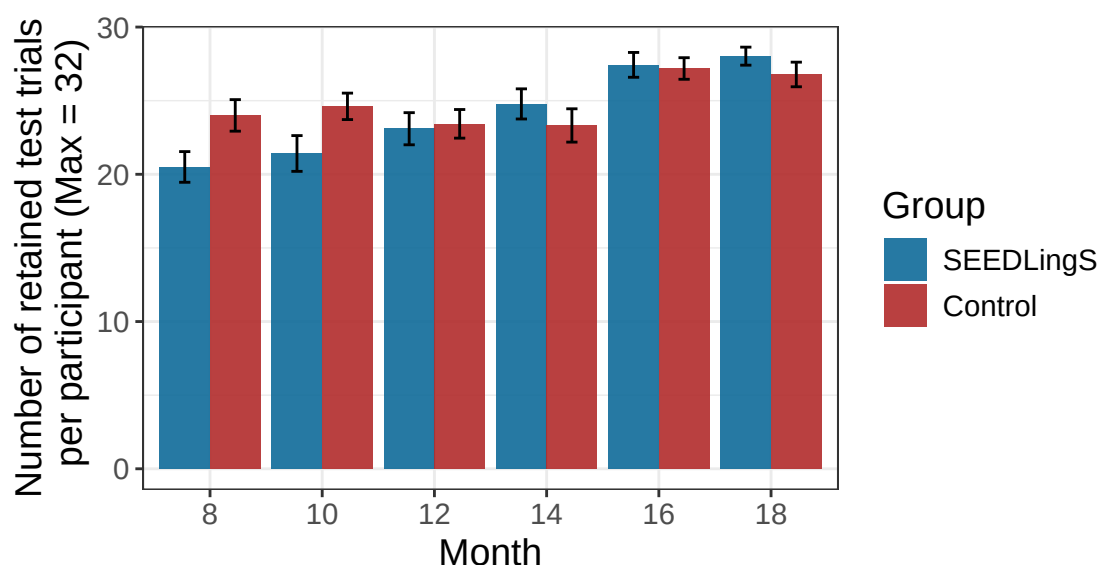


Figure 7

Distribution of the number of retained trials contributed per participant among SEEDLingS and control participants over months. Vertical lines indicate standard errors. Overall, the two samples contributed a similar number of usable trials per participant.

3.2.3.1 Amount of Retained Data. We first compared the two groups on amount of retained data. As seen in Fig. 7 and confirmed via linear regression, older infants contributed more data than younger infants but this did not vary by sample. see the Supplementary Materials for details. We thus found no evidence that SEEDLingS participants contributed more usable trials than controls, suggesting a limited role for repeated testing experience.

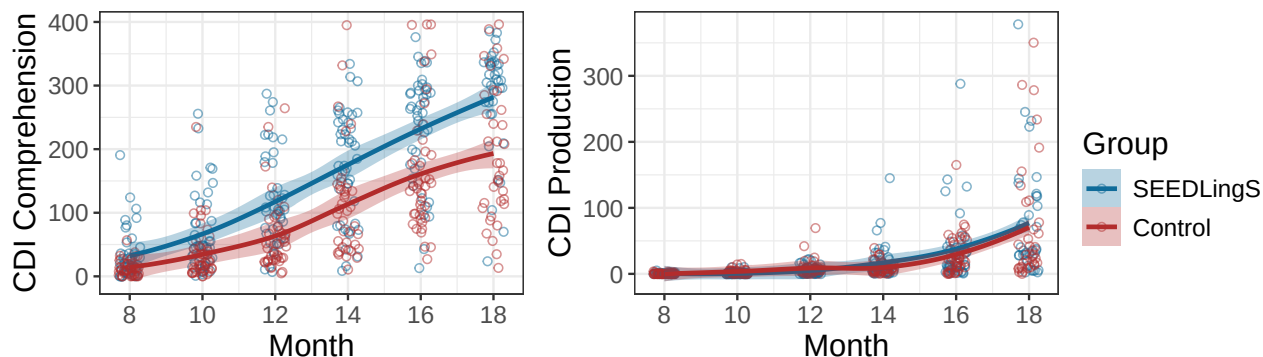


Figure 8

CDI comprehension (left) and production scores (right) for SEEDLingS and control participants, by age. Circles show individual infants' CDI scores at a given month. LOESS regression lines and corresponding 95% CI are in blue (SEEDLingS) or red (control). While CDI production scores were comparable between groups, SEEDLingS participants showed higher CDI comprehension scores over time.

3.2.3.2 Vocabulary Size.

We next compared the two groups on both receptive and productive vocabulary size, as measured by CDI (see Fig. 8). Parents of SEEDLingS participants filled out the CDI survey for 249 out of 264 eyetracking visits (6 planned per participant), and parents of control participants completed 242 surveys. Linear regression predicting CDI comprehension scores found a main effect of age, group, and an age-by-group interaction, while production scores fit with the same terms only showed a main effect of age (see the Supplementary Materials for details).

In sum, parent-reported CDI data showed that while both groups showed similar growth in productive vocabulary, SEEDLingS participants had larger receptive vocabulary overall and showed greater increases over age compared to control participants. Therefore, we controlled for overall receptive vocabulary size in subsequent between-group comparisons of eyetracking performance.⁹

⁹ The Supplementary Materials detail reasons to interpret the group difference in CDI comprehension scores with caution (e.g., due to variation in how parents of the longitudinal vs. cross-sectional samples

3.2.3.3 *Word Comprehension.*

In this final set of analyses, we compare SEEDLingS and control infants' performance across the three trial types. The comparison on generic items served as a simple check, as we did not expect any group differences: These items consisted of a predetermined list of words paired with novel, prototypical images and should be similar across groups in both frequency and familiarity. In contrast, audio-nouns were high-frequency words selected from SEEDLingS infants' audio recordings, paired with novel, prototypical images. Because these words were more frequent for SEEDLingS infants, we predicted better performance in this group compared to controls. Video-nouns were high-frequency words paired with familiar images drawn from SEEDLingS infants' video recordings. We expected an even greater advantage for SEEDLingS infants on video-nouns, since these items were both more frequent and more familiar for them. For each trial type, we first compare group performance over time by fitting a full linear mixed-effects model predicting the increase in target looking as a function of participant group, age, and their interaction (full model: `increase in target looking ~ trial type × age-scaled + (1|subject) + (1|test pair)`). We then assess whether any observed group differences persist after accounting for receptive vocabulary size.

Generic Items.

3.2.3.3.1 Contrary to our prediction, we did observe group differences on generic items (Fig. 9 left). The best-fitting model retained all fixed effects and revealed a significant main effect of age ($b = 0.05$, $SE = 0.01$, $t(23.66) = 7.85$, $p < .001$), no significant main effect of participant group ($b_{\text{SEEDLingS}} = -0.01$, $SE = 0.02$, $t(121.46) = -0.34$, $p = 0.73$), and a significant age $\times$ group interaction ($b_{\text{SEEDLingS:age}} = 0.02$, $SE = 0.01$, $t(1,103.25) = 2.22$, $p = 0.03$). This interaction shows that SEEDLingS participants improved more in word comprehension over time compared to control participants. Specifically, for each two-month increase in age, SEEDLingS participants' comprehension of generic words increased by 0.07 units, but control participants' only

filled out the CDI logistically, and because SEEDLingS parents might be more sensitive to their child's vocabulary knowledge)

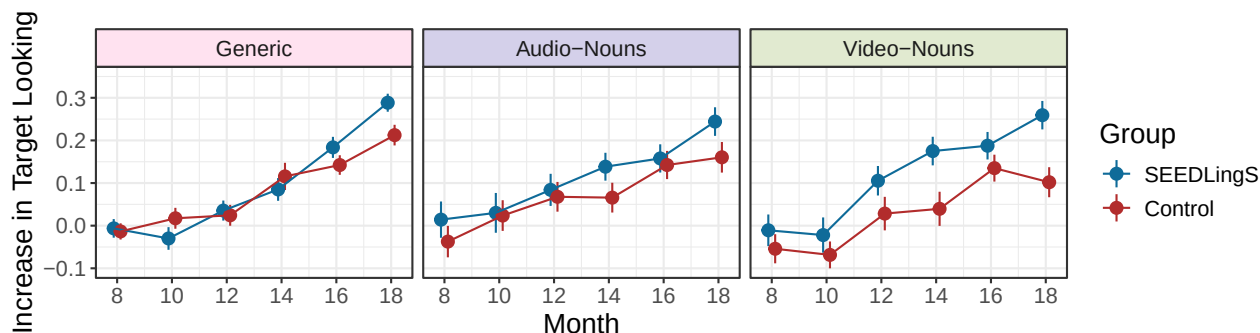


Figure 9

Word comprehension (operationalized as increase in target looking) on each trial type (generic, audio-nouns, video-nouns) by participant group (SEEDLingS, control) over months. Dots represent means, and vertical lines indicate standard errors. SEEDLingS participants improved more than controls over time on generic items and consistently outperformed controls on audio-nouns and video-nouns.

increased by 0.05 units.

Given the high correlation between CDI comprehension and age ($r_{\text{SEEDLingS}} = 0.77$, $r_{\text{control}} = 0.66$), we did not directly add CDI comprehension as a covariate to the age $\times$ group model above due to concerns about multicollinearity. Instead, we focused on performance at 18 months—the age at which the group difference in comprehension appeared largest—and added CDI comprehension as a covariate in a separate cross-sectional analysis. This approach required excluding additional participants, because at 18 months, only 36 out of the 44 SEEDLingS participants and 34 out of the 41 control participants retained for eyetracking analyses had CDI data.

As detailed in the Supplementary Materials, the group difference in comprehension of generic items was not robust, at least not at 18 months. Among the subset of participants with CDI data, participant group did not significantly predict eyetracking performance, even before controlling for CDI comprehension. In contrast, individual variation in CDI comprehension significantly predicted eyetracking performance.

Audio-Nouns. 3.2.3.3.2 Next, we compared the two groups on audio-noun comprehension (Fig. 9 middle). Model selection removed the age $\times$ group interaction and all random effects, resulting in a linear model. The final model revealed a significant main effect of age ($b = 0.04$, $SE = 0.01$, $t(832) = 6.89$, $p < .001$) and participant group ($b_{\text{SEEDLingS}} = 0.04$, $SE = 0.02$, $t(832) = 2.02$, $p = 0.04$). The significant main effect of participant group (with stronger comprehension for SEEDlingS than Controls), in the absence of a significant interaction, suggests that SEEDlingS participants were overall better at comprehending audio-nouns (which were high-frequency nouns selected from their home environments) than control participants were.

However, as detailed in the Supplementary Materials, the observed group difference in audio-noun comprehension found in the full sample was not statistically reliable in the reduced sample of participants with CDI data. That said, CDI comprehension also did not reliably predict audio-noun comprehension. This creates an interpretive challenge: Subsetting each sample to the ~82% that had CDI data removed the group effect on audio-nouns comprehension, but CDI scores did not account for variance of these data either.

Video-Nouns. 3.2.3.3.3 Finally, we compared the two groups on video-noun comprehension (Fig. 9 right). Model selection removed the age $\times$ group interaction and the random intercept for test items. The final model revealed a significant main effect of age ($b = 0.05$, $SE = 0.01$, $t(874.50) = 8.49$, $p < .001$) and participant group ($b_{\text{SEEDLingS}} = 0.08$, $SE = 0.02$, $t(874.50) = 3.48$, $p < .001$). The significant main effect of participant group (SEEDlings $>$ Controls), in the absence of a significant interaction, suggests that SEEDlingS participants were overall better at comprehending video-nouns (which were high-frequency nouns paired with familiar referents from their home environments) than control participants.

This time, as detailed in the Supplementary Materials, the group difference in

video-noun comprehension observed in the full sample was still statistically significant in the reduced sample of participants with CDI data, even controlling for CDI comprehension. CDI comprehension itself was not a significant predictor in the reduced sample.

In summary, our between-group analyses suggest a minimal influence of repeated testing: SEEDLingS infants did not contribute more usable data than control infants. The two groups differed in receptive, but not productive, vocabulary, which we accounted for in subsequent comparisons of word comprehension. While group differences emerged across all trial types, only the effect observed for video-nouns (high-frequency words paired with familiar referents for SEEDLingS participants) remained robust in analyses controlling for vocabulary. In contrast, the effect observed for audio-nouns (high-frequency words paired with novel referents for SEEDLingS participants) was less robust. Together, these findings offer suggestive evidence for the role of word frequency in word comprehension and convergent evidence that infants have more difficulty comprehending words when they are paired with novel referents drawn from others children’s personal experiences.

4 Discussion

Across two studies, we examined how infants’ everyday experiences with specific words and referents shape early word comprehension. By combining naturalistic home recordings with in-lab eyetracking assessments, we directly tested how word frequency and referent familiarity influence infants’ understanding of concrete nouns between 8-18 months. We used a novel approach that involved hand-tailoring each eyetracking experiment to an individual infant’s experience: For each longitudinal SEEDLingS infant, we hand-picked high-frequency nouns from their own recent home recordings and paired them either with novel, prototypical referents (audio-nouns) or with familiar referents seen in their home (video-nouns). These hand-picked items were compared against a set of common but lower-frequency generic nouns paired with prototypical but not child-specific referents (generic). A cross-sectional control group, matched in age and gender but less

familiar with the hand-picked stimuli, completed the same tasks to provide a benchmark for interpreting the role of word- and referent-specific experience.

We found two key results. First, word frequency reliably predicted infants' word comprehension, but only during a narrow developmental window: Between 12-14 months, SEEDLingS infants showed better comprehension for higher-frequency audio- and video-nouns compared to lower-frequency generic nouns (though note even generic nouns were relatively high frequency). This effect was further supported by finer-grained analyses using continuous frequency estimates showing that more frequent words in the home were more likely to be recognized in the lab. Together, these findings suggest that the relationship between frequency and comprehension changed nonlinearly with age, with sensitivity to frequency peaking at 12-14 months. Second, we found little evidence that referent familiarity provided an additional boost to comprehension. SEEDLingS infants' performance was similar for audio-nouns (paired with novel, prototypical referent images) and video-nouns (paired with familiar referent images), suggesting that infants generalized their understanding of common nouns across referents. However, control infants, who had neither particularly high exposure to these words nor familiarity with the referents, showed reduced performance on video-noun trials, indicating that the idiosyncratic referents used in those trials were more difficult to recognize. Between-group comparisons confirmed that the patterns observed in SEEDLingS infants were not simply due to repeated testing, as control infants showed similar developmental trajectories on generic items after accounting for differences in receptive vocabulary. These comparisons also provided additional support for the role of experience: SEEDLingS infants outperformed controls on hand-picked items, with robust advantages for familiar referents and suggestive evidence for word frequency effects.

We also observed an unexpected group difference in how well infants understood our generic items, i.e., common nouns depicted with prototypical referent images, of the kind

standardly used in prior work. We had expected similar developmental trajectories across groups, i.e., a nonlinear improvement over development, as observed in prior studies (Bergelson, 2020). However, only SEEDLingS infants showed this pattern. That said, these unexpected group differences were not particularly robust: They were dropped from model selection and not significant when included alongside CDI comprehension scores (within the majority of kids for whom those were available; see the Supplementary Materials). We limit speculation beyond noting the possibility that the experiment had higher processing load for controls, given the mix of prototypical and child-specific items selected based on another child.

In the remainder of the discussion, we discuss the implications of our results for how frequency and referent familiarity shape early comprehension. We begin by discussing the role of word frequency, elaborating on the nonlinear shift in frequency effects on comprehension over development. We consider how this pattern aligns with theories of “learning through processing” (Meylan & Bergelson, 2022) and reflect on how our outcome measure (word comprehension via the looking-while-listening paradigm) might be complemented by future work targeting real-time processing efficiency. Next, we discuss infants’ ability to generalize across exemplars and what the observed difficulties with others’ hand-picked referents suggest about early lexical representations and category learning. We close by considering how these results inform broader questions about category learning, semantic representation, and developmental change in early word understanding.

4.1 Interpreting Real-Time Effects of Frequency

We observed a nonlinearity in how the effect of word frequency on SEEDLingS infants’ word comprehension shifts with age. At 8-10 months, performance was at chance across the board, making it difficult to detect frequency effects. While prior studies have detected small but consistent effects that infants begin to comprehend some common nouns

by this age (Bergelson & Swingley, 2012; Foushee & Srinivasan, 2024; Kartushina & Mayor, 2019; Tincoff & Jusczyk, 1999, 2012), our experiment may have dampened early effects by testing each infant on a unique, hand-tailored set of items. Unlike previous studies that tested all infants on the same set of high-frequency words, this approach likely introduced variability across participants, making it harder to detect the relatively modest early comprehension effects reported in prior work. It is also possible that our efforts to pick “good” generic items (i.e., common nouns based on CHILDES and prototypical referent images) were suboptimal for eliciting word knowledge from these young infants (either because as a group they do not know these words, or because our prototypical images were not as recognizable as intended; all stimuli are on osf.io/m2kdz for the curious). We therefore focus our discussion on the developmental window between 12-18 months, where infants began to show above-chance comprehension.

As visualized in Fig. 6, at 12 and 14 months, SEEDLingS infants understood more frequently heard words better than less frequently heard ones. By 16-18 months, however, infants performed above chance regardless of frequency. This developmental shift suggests that while early learning might be constrained by frequency of exposure, infants become more efficient learners over time, gradually relying less on input frequency alone. They are better able to learn lower-frequency words, likely due to gains in language processing efficiency. This shift aligns with the “learning through processing” framework (Meylan & Bergelson, 2022), which characterizes word learning as an iterative process: Infants use their current language knowledge to interpret input in context, then update that knowledge based on the inferred meaning. The updated knowledge, in turn, shapes future processing, creating a feedback loop in which learning and processing continuously influence one another.

Past work shows that the linguistic input available to infants changes little over time, across measures like word counts, number of speakers, and utterance types

(Bergelson et al., 2018; Bergelson, 2020). In line with this characterization, our supplementary analyses further suggest that within-child estimates of test-word frequency are highly correlated across temporal windows (two preceding months, all preceding months, or all months). This is also consistent with prior proposals (e.g., Meylan & Bergelson, 2022) that developmental shifts in comprehension likely reflect changes in how input is processed and integrated into existing knowledge rather than changes in input itself. Indeed, older infants can draw on a broader set of learning signals, which may contribute to their improved data processing efficiency. For example, at around 12-14 months, infants begin to comprehend social-pragmatic cues such as the pointing gestures (Behne et al., 2012; Carpenter et al., 1998). By around 15 months, they also begin to exploit morphosyntactic information to constrain hypotheses about novel word meanings, a process known as syntactic bootstrapping (for reviews, see Babineau et al., 2024; Fisher et al., 2010). Our findings suggest that frequency effects are not static but emerge dynamically from changes in learning mechanisms over time.

In this study, we used word frequency to predict word comprehension outcomes, assessed through looking-while-listening eyetracking experiments. This method's main strength is in providing an objective measure of word comprehension in infancy. However, the way we analyzed the eyetracking data does not directly assess how *efficiently* infants process known words in real time. Specifically, our primary analyses focused on the proportion of target looking within a 3-second window following target word onset, without examining the temporal dynamics of gaze behavior within that window (though we note that reaction time and target looking accuracy are of course closely linked). In adults (who have mature lexical knowledge), more frequent words are typically processed more efficiently (e.g., Dahan et al., 2001; Magnuson et al., 2007). Whether similar frequency-based processing advantages are present in infancy remains an open question. Future work could examine reaction time and the shape of the timecourse of target looking to assess how processing efficiency changes as words are acquired. While this is beyond the

scope of the present study, we make our dataset available at osf.io/m2kdz and invite researchers to explore these dynamics. Visualizations of the timecourse data are provided in the Supplementary Materials.

Importantly, frequency effects observed in real-time processing may themselves reflect learning history. On this view, faster processing of high-frequency words is not a separate phenomenon but a downstream consequence of earlier exposure-driven word learning. Prior accounts have attributed frequency-based processing advantages to stronger lexical connections or higher resting activation levels in the mental lexical (as reviewed by Brysbaert et al., 2018). That is, what appears as a processing advantage may simply reflect more robust lexical representations built through prior learning experience. This interpretation reinforces the idea that learning and processing are not distinct stages but deeply intertwined during word learning. This is again consistent with learning-through-processing, whereby processing efficiency both results from and supports learning (Meylan & Bergelson, 2022).

4.2 From Familiar Referents to Robust Categories

Our findings suggest that SEEDLingS infants were able to extend words not only to familiar referents from their homes, but also to novel, prototypical exemplars. This pattern supports the idea that from the earliest stages of word learning, infants map common nouns to broad conceptual categories rather than a specific category member, consistent with the broad-to-narrow or broad-and-narrow views of lexical development outlined above.

In contrast, control infants struggled with unfamiliar referents drawn from other infants' idiosyncratic experiences. One possible explanation is that infants' early lexical representations are overly narrow, reducing their ability to recognize unfamiliar category members as instances of words they know. An alternative explanation is that infants do form appropriately scoped conceptual categories, but their word recognition is influenced by other properties of the referents. Because the video-noun referents were selected to

match particular infants' recent experiences, they were more idiosyncratic, and thus sometimes less prototypical, or more dependent on local communicative context than the standard lab-supplied images used in the other conditions. For example, while most baby shoes would be readily recognized as shoes, a particular lego-set toy fish is certainly not as typical of a fish as a goldfish or trout is to most people and may therefore be harder to recognize (Rosch et al., 1976). For some nouns such as "water", a pink bottle can serve as the intended referent if the local communicative context makes clear that it contains water (i.e., if that is the sippy cup used for water for a given child), but an image of water itself may have been more transparent out of context.

As noted in Study 2 and detailed in the Supplementary Materials, adults also showed greater difficulty on video-noun trials than on audio-noun or generic trials when tested on the same image pairs used in the infant experiments. This suggests that control infants' struggle with video-nouns may reflect a processing challenge posed by the particular child-specific referents rather than overly narrowly scoped word meanings. For this reason, the control-group pattern is informative about the relative difficulty of the video-noun stimuli, but less directly diagnostic of the scope of infants' lexical representations.

The same consideration is relevant for interpreting the SEEDLingS pattern. SEEDLingS infants performed similarly on audio- and video-noun trials, suggesting that their noun knowledge was not limited to familiar referents from their own recent experience. However, if video-noun stimuli were more idiosyncratic or harder to identify out of context, then SEEDLingS infants' familiarity with those referents may have helped offset the additional processing demands. Thus, the audio- vs. video-noun comparison is relevant to the question of whether infants extend nouns beyond familiar referents, but it does not isolate the scope of semantic representations from differences in stimulus typicality, identifiability, or the degree of contextual inference required to identify the

intended referent. The results are therefore consistent with appropriately scoped semantic representations, while leaving open the possibility that referent familiarity and stimulus idiosyncrasy jointly shaped performance.

More broadly, the present study addresses one aspect of the broad vs. narrow question by testing extension from familiar referents to less familiar ones. At the same time, it does not directly address how robust infants' lexical representations are across variation in referent typicality (see also Weaver et al., 2024, who report limited evidence for typicality effects in infant word comprehension), or how infants integrate contextual information to identify the intended noun referent. Future work could address this issue more directly by systematically manipulating both factors.

Relatedly, across our two samples of nearly 300 infants spanning two regions of the U.S., the prevalence of certain nouns (and their various specific referents) was both wide-ranging and relatively conserved (see Fig. 3). As the demographics of our samples show, our study involved two relatively homogeneous sets of children (largely white children from families with a high degree of maternal education, all predominantly learning English). This homogeneity likely increased the likelihood that certain toys, books, foods, and therefore *nouns* were common across them. Past work shows that children's input can differ across languages and cultures. For example, the relative frequency and salience of nouns and verbs differs between Korean- and English-learning children (Au et al., 1994; Choi & Gopnik, 1995), and how infants and caregivers interact with objects can vary across communities (e.g., in the U.S. and in Vanuatu, Little et al., 2016). To our knowledge, however, there is little directly comparable cross-linguistic or cross-cultural work testing infants' word comprehension when given familiar vs. unfamiliar referents. Further work from wider ranging samples would be a welcome addition for better understanding word- and referent-level variability and its potential effects on learning.

4.3 Learning Category Representations

Our findings raise new questions about how infants form and represent early categories. If familiarity with specific referents is not necessary for successful word comprehension, at least for SEEDLingS infants, then what kinds of early experiences support the formation of broad categories, whether semantic, conceptual, or perceptual? One possibility is that infants learn to abstract across referents by identifying diagnostic perceptual (e.g., shape, Althaus & Plunkett, 2016) or conceptual features (e.g., object function, Träuble & Pauen, 2007) that indicate a category. Various factors may help highlight these features. One example is exemplar variability: Prior work has shown that exposure to a range of exemplars facilitates category learning by drawing infants' attention to the consistent features shared across instances (e.g., Perry et al., 2010; Twomey et al., 2014; Vukatana et al., 2015). Infants may also be guided by an early bias to interpret words as referring to categories, rather than individuals (Waxman & Markow, 1995), which can direct their attention to commonalities across potential category members (Althaus & Plunkett, 2016).

To better test these possibilities, we need richer documentation of the properties of exemplars infants encounter in their everyday lives (Clerkin & Smith, 2022; Long et al., 2024; Pereira et al., 2010; Sullivan et al., 2021). Many studies on category learning use novel words and artificial categories in tightly controlled lab settings. While useful, these paradigms may not fully capture how infants acquire natural categories in everyday life. One particularly promising dimension for future research is within-category perceptual variability, given how much more inherently variable some real-world categories are than others. For example, dogs vary more in shape, size, and color than cats do, making 'dog' a more visually variable category. Prior work has shown that such variability can influence infants' category boundaries: When exposed to a more visually diverse set of dogs, infants are more likely to overextend the category to include cats (Quinn et al., 1993). In general,

categories with higher internal variability may need more observations to learn (Fried & Holyoak, 1984), but may also support better generalization to novel exemplars (Hahn et al., 2005).

Critically, we can characterize within-category variability along two dimensions: the amount of inherent variation within a category (e.g., how different dogs look from one another) and the diversity of exemplars an individual infant actually encounters (e.g., seeing only one small white dog vs. seeing dogs of many breeds and sizes; for a review, see Raviv et al., 2022). These two dimensions of variability likely interact in shaping category representations. A child exposed to a visually variable category through only a narrow subset of exemplars may develop a more limited or skewed representation than a child who experiences a broader sample of the category. Future work that carefully analyzes infants' visual environments (e.g., Long et al., 2024; Pereira et al., 2010) could help disentangle these contributions and shed light on how infants construct categories from their everyday experience. It will also be valuable to integrate linguistic distributional information (e.g., how often and in what contexts concrete nouns appear; Clerkin & Smith, 2022; Sullivan et al., 2021) and to build computational models that align visual and linguistic input. Such cross-modal models could illuminate how early category and noun learning are shaped by the structure of both perceptual and linguistic experience (Aho et al., 2023).

5 Conclusion

Taken together, this work highlights the role of infants' early experiences in shaping their word learning, leveraging hand-tailored eyetracking experiments informed by naturalistic home recordings. Infants' real-time word comprehension not only improved with age, but also showed sensitivity to word frequency, particularly around their first birthday; word comprehension was uniformly strong across frequency levels later in the second year of life. This pattern is consistent with a mechanistic shift in how infants process linguistic input, becoming more efficient learners over time. Infants extended

common nouns equally well to both familiar and novel exemplars, indicating appropriately scoped semantic representations from the earliest stages of word learning. These findings (and the rich data supporting them) open the door for future research to further examine how variability in referent properties and linguistic information interact to shape infants' emerging word knowledge.

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¹⁴¹⁵ Zhu, H. (2024). *kableExtra: Construct complex table with 'kable' and pipe syntax*.

Supplementary Materials

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The sections below contain the Supplementary Materials for the main manuscript.

1418

A. Computing and Comparing Frequency

1419

B. Exclusion Criteria

1420

C. R Packages

1421

D. More Details on Comparing SEEDLingS and Control Participants

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E. Adult Control Study

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F. Timecourse Visualizations

Appendix A

Computing and Comparing Frequency

In this section, we present more details on how we computed the relative frequency measure and compared frequency across trial types within each participant group.

7.1 Annotating Noun Tokens in Recordings

We annotated the concrete noun tokens from infants' home recordings according to the following steps (Kalenkovich et al., 2025). First, each token was transcribed exactly as spoken (e.g., including affixes, compounds, diminutives, wordplay, etc.). To group related forms within a recording, each noun was then assigned a *basic level* form, defined as the most common version of that noun in that recording. These basic level forms were used in the eyetracking sessions. For example, if a family said 'bear', 'bears', and 'teddy bear', and 'teddy bear' appeared most frequently (with 'bear' in any form also ranking among the top nouns), then 'teddy bear' was selected as the test word. To ensure consistency across families, we then standardized all basic level nouns throughout the SEEDLingS corpus by assigning each one a *global basic label*, which functioned like a lemma (Moore & Bergelson, 2024). For instance, tokens labeled 'bear' or 'teddy bear' at different timepoints were consolidated under the global basic level 'bear'. To compare word frequencies across infants, each test word was converted to its global basic level. For example, if infant X was tested on 'teddy bear', the frequency of that word was calculated by counting all noun tokens with the global basic level 'bear', and dividing that by the total number of noun tokens in their recordings from the two months prior to the eyetracking session. Note that a highly similar procedure for computing frequency has been used in other work with the broader SEEDLingS dataset (e.g., Dong et al., 2026), so this description overlaps somewhat with others.

Table A1

Mean and standard error of untransformed test pair frequency across trial types (generic, audio-nouns, video-nouns) within each participant group at each month.

Participant Group	Month	Generic	Audio-Nouns	Video-Nouns
SEEDLingS	8	0.57% (0.080%)	2.02% (0.178%)	2.14% (0.174%)
SEEDLingS	10	0.66% (0.054%)	1.92% (0.119%)	1.72% (0.122%)
SEEDLingS	12	0.60% (0.053%)	1.60% (0.107%)	1.72% (0.150%)
SEEDLingS	14	0.69% (0.051%)	1.67% (0.126%)	1.75% (0.134%)
SEEDLingS	16	0.47% (0.036%)	2.01% (0.145%)	1.60% (0.108%)
SEEDLingS	18	0.45% (0.028%)	1.46% (0.076%)	1.33% (0.080%)
Control (Proxy Estimates)	8	0.57% (0.025%)	0.93% (0.051%)	0.72% (0.060%)
Control (Proxy Estimates)	10	0.66% (0.019%)	0.82% (0.055%)	0.57% (0.047%)
Control (Proxy Estimates)	12	0.63% (0.019%)	0.62% (0.044%)	0.58% (0.056%)
Control (Proxy Estimates)	14	0.68% (0.027%)	0.66% (0.041%)	0.48% (0.042%)
Control (Proxy Estimates)	16	0.46% (0.019%)	0.70% (0.045%)	0.55% (0.048%)
Control (Proxy Estimates)	18	0.44% (0.013%)	0.43% (0.031%)	0.45% (0.038%)

7.2 Comparing Frequency Between Trial Types

Raw word frequencies are known to be highly skewed (e.g., following Zipfian distributions), and it is standard practice in psycholinguistics to apply a log transformation before entering frequency as a dependent variable in statistical models. However, as shown in Fig. 3, the distributions of log-transformed test pair frequencies remain visibly non-normal. This is particularly true for the proxy frequency estimates derived for control infants based on SEEDLingS infants' recordings, where the distributions are not even approximately bell-shaped. Therefore, we adopted nonparametric methods in this section to analyze untransformed test pair frequency. See Table A1 for summary statistics by participant group, month, and test pair type.

We first conducted Kruskal-Wallis tests (the nonparametric equivalent of one-way ANOVA) to examine whether frequencies differed across the three trial types (generic, audio-nouns, and video-nouns) within each participant group at each month (see Table A2). For the SEEDLingS group, we observed significant differences across trial types at all timepoints between 8-18 months ($ps < .001$). This pattern was expected given that the audio-nouns and video-nouns were selected based on their high frequency in SEEDLingS infants' recordings, whereas the generic words were not.

For the control group, the pattern was more nuanced. We again found significant differences found at all timepoints ($p < 0.01$) except at 18 months ($p = 0.22$). However, the test statistics for control infants were markedly smaller (χ^2 s ranging from 9 to 29) compared to SEEDLingS infants (χ^2 s > 100), indicating that even when differences existed among trial types, they were less pronounced in the control group.

To further explore which pairs of trial types differed in frequency, we conducted pairwise Wilcoxon Rank Sum Tests (nonparametric equivalent of two-sample t-tests) with Bonferroni corrections (Table A3). Results for the SEEDLingS group again aligned with our experimental design: For all months, both generic vs. audio-nouns and generic vs. video-nouns comparisons were highly significant ($ps < .001$), while audio- vs. video-nouns were statistically indistinguishable ($ps = 1$), with a single exception at 16 months ($p = 0.02$). However, even this exception had a higher p -value than the other two comparisons at 16 months, suggesting that the audio- and video-nouns were still more similar to each other than either was to the generic words.

By contrast, the control group exhibited no consistent pattern of differences in frequency. Some pairwise comparisons reached significance, but the effects were inconsistent across months and trial types. Importantly, when differences were detected, they were associated with higher p -values and smaller effect sizes than in the SEEDLingS group.

Together, these findings indicate that our frequency manipulation was effective and reliable for the SEEDLingS infants: Audio- and video-nouns were consistently more frequent than generic nouns, but not significantly different from each other. In contrast, the same high-frequency stimuli were not systematically more or less frequent for the control group.

7.3 Frequency Estimates Across Temporal Windows and Aggregation Levels

We next evaluate whether our findings with SEEDLingS infants depend on how word frequency is operationalized. We first vary the **temporal window** over which frequency is calculated. In the main analyses, frequency was computed based on corpus data from the *recent past* (i.e., the two preceding months). Here, we additionally consider the *full history* (all preceding months) and the *entire corpus* (all months). For example, to compute frequency at test month 14, the recent past considers corpus data at 12-13 months, the full history includes 6-13 months, and the entire corpus involves all data between 6-17 months.

We also vary the level of aggregation across infants. Whereas the main analyses computed frequency separately for each infant, we also consider frequency computed by *pooling across all infants*. Crossing these factors yielded five alternative frequency measures (excluding the one reported in the main text based on individual infants' recent past). For each measure, we assessed a) its correlation with the original frequency metric, and b) its ability to predict SEEDLingS infants' eyetracking performance.

We quantified the correlation between each alternative frequency measure and the original measure at both the level of individual test words and test pairs. We used three complementary approaches: 1) Pearson correlations between log-transformed frequencies (excluding zero-frequency items), 2) Spearman non-parametric correlations on untransformed frequencies, and 3) linear mixed-effects models of the form $\text{LogOriginalFreq} \sim \text{LogAltFreq} + (1|\text{Subj}) + (1|\text{Month})$ (again excluding

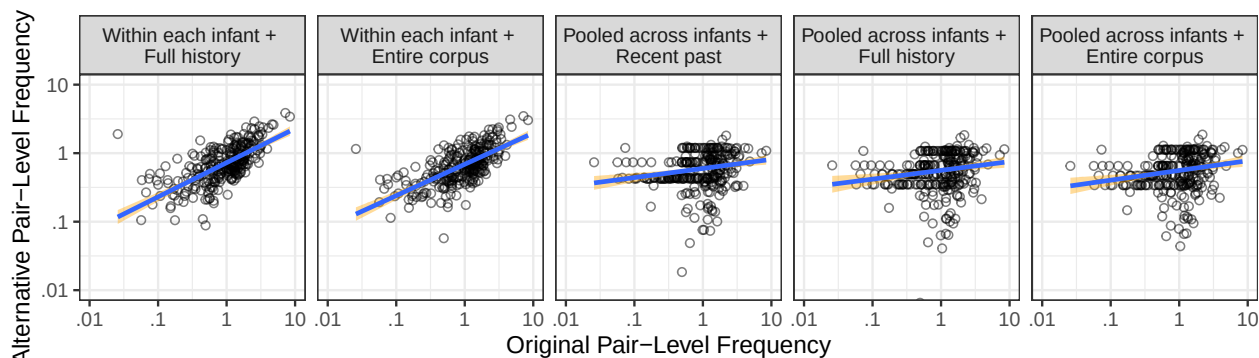


Figure A1

Pair-level frequency estimates at 14 months under five alternative operationalizations, compared with the original within-child recent-past measure. Each point represents a test pair. Frequencies are shown on log scales after excluding zero-frequency pairs (~3% of pairs). Robust linear regression lines are in blue, 95% CIs in orange. Within-child frequency measures are highly correlated with the original measure, whereas pooled frequency measures show weaker correlation

zero-frequency items).¹⁰ The mixed-effects models account for repeated observations within infant and test month. The fixed-effect coefficient for `LogAltFreq` provides an estimate of the correlation between log-transformed frequency measures (similar to Pearson), while controlling for the nested structure of the data.¹¹

We found that when frequency was computed separately for each infant, alternative measures were highly correlated with the original measure (correlations $\approx 0.6 - 1.0$), regardless of the temporal window. In contrast, when frequency was pooled across infants, correlations with the original measure was substantially lower across temporal windows

¹⁰ 14% of test words and 3% of test pairs had zero frequency and were excluded when computing the Pearson correlations and regression coefficients.

¹¹ For comparisons involving the original metric and the full-history within-child metric, month 8 was excluded because the two measures yield identical frequency estimates at this time point, as both are based on each infant's corpus data at 6-7 months.

(correlations $\approx 0.2 - 0.5$). This pattern was similar at both the word and pair levels, with slightly higher correlations for pair-level estimates. Results are summarized in Table A4. We further visualize these correlation patterns in Figure A1 using data from 14 months as an illustrative example. Consistent with the correlation analyses, frequency measures computed within each infant show strong alignment with the original measure, whereas the pooled measures show weaker correspondence and greater dispersion. Notably, the pooled plots show horizontal bands because the generic test pairs are shared across infants; when frequency is computed across all infants, these shared pairs receive identical frequency estimates, resulting in repeated values along the y-axis.

We next evaluated how well each alternative frequency measure predicted word comprehension. As in Section 2.2.3, we fit linear mixed-effects models predicting increase in target looking from log-transformed test pair frequency, linear and quadratic age, and their interactions.

When frequency was computed separately for each infant, the alternative measures yielded patterns comparable to the main analyses (see Tables A5 and A6). Model selection retained all fixed effects in both cases (final model: `increase in target looking ~ log test pair frequency × (age-scaled + age-scaled2) + (1 | subject)`). In both alternative operationalizations, the relationship between word comprehension and frequency varied nonlinearly with age: frequency effects were strongest at intermediate ages (12–14 months) and weaker at younger and older ages. Estimated effect of frequency on word comprehension by age are summarized in Table A7.

In contrast, when frequency was pooled across infants, it was a poor predictor of word comprehension, regardless of included age range. All fixed effects involving frequency were excluded during model selection.

To summarize, across a range of alternative operationalizations of word frequency,

our results are robust to the choice of temporal window (i.e., whether frequency is computed over the recent past, full history, or entire corpus). In contrast, the results vary by the aggregation level (i.e., whether frequency is computed within each infant or pooled across infants). Frequency measures computed within infants are highly correlated with one another and reliably predict word comprehension, whereas pooled frequency measures show weaker correlation with the original frequency measure and limited predictive value. These findings support our analytic choice in the main text to compute frequency separately within each infant, and indicate that restricting frequency to a recent temporal window does not qualitatively alter the observed relationship between frequency and word comprehension.

7.4 Representativeness of Highest-Talk Audio Segments

In this section, we assess whether frequency estimates derived from the highest-talk audio segments are representative of those obtained from the full daylong recordings. For infants aged 6-7 months, the full daylong audio recordings were manually annotated, whereas for older ages (8-17 months) annotations were restricted to the highest-talk segments (top 4 hours for 8-13 months; top 3 hours for 14-17 months) due to logistical constraints. We therefore use the 6-7 month corpus (for which full daylong annotations are available) to directly compare estimates from highest-talk segments with those from full daylong recordings.

We focused on the set of test words used at 8 months (across all trial types: generic, audio-nouns, video-nouns) and computed word frequency from the two preceding months (6-7 months). This set forms a reasonable sample of nouns for our purposes and, by design, spans a range of frequencies: audio-nouns were specifically selected to be highly frequent in the audio recordings, generic nouns were not selected based on frequency at an individual level, and video-nouns were selected based on their frequency in the video recordings. Because video-nouns were reliably observed in infants' home environments (via video

recordings), they are also likely to occur in the audio recordings, whereas no such assumption applies to generic nouns. Accordingly, in the audio-only transcripts we expect audio-nouns to be more frequent than generic nouns, and both audio- and video-nouns to be more frequent than generic nouns overall, though we had no strong a priori prediction about the ordering of audio- vs. video-nouns in the audio transcripts alone.

For each test word, we computed frequency in two ways: (1) using only the top 3 highest-talk hours per child and (2) using all hours of the daylong recordings. We then compared the resulting frequency distributions.

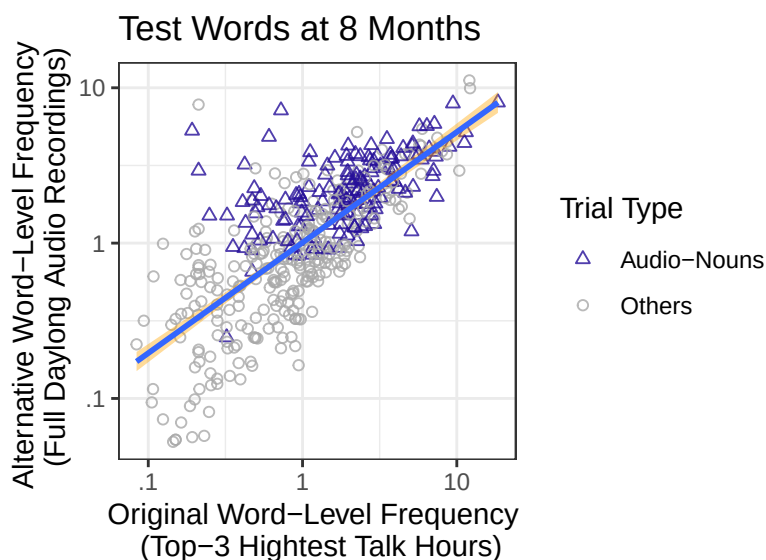


Figure A2

Representativeness of highest-talk audio segments for 8-month test words. For each test word used at 8 months, frequency was computed from the 6-7 month corpus in two ways: using only the top-3 highest-talk hours (as reported in the main text; x-axis) or using the full daylong recordings (y-axis). Each point represents one test word. Audio-nouns are shown as blue triangles; all other test words (video-nouns and generic nouns) are shown as gray circles. Robust linear regression line is in blue, 95% CI in orange. Frequencies derived from the two sampling approaches were highly correlated, and audio-nouns occupied the high-frequency end of the distribution under both measures.

We found that frequency estimates derived from the top-3 hours and from the full daylong recordings were highly correlated (Pearson correlation on log-transformed frequencies, excluding zero-frequency items: $r_p = 0.74$; Spearman rank correlation on raw frequencies: $r_s = 0.78$), indicating that estimates are stable with respect to the amount of audio recordings included. As shown in Figure A2, the two measures show a strong linear relationship across test words.

In addition, the relative ordering of trial types was preserved across both sampling approaches. Audio-nouns (and video-nouns) were significantly more frequent than generic nouns in the audio transcripts regardless of whether frequencies were computed from the top-3 hours or from the full daylong recordings (Wilcoxon Rank-Sum tests with Bonferroni correction, $ps < .001$). Consistent with this pattern, audio-nouns occupy the high-frequency end of the distribution under both sampling schemes (Figure A2), confirming that these items were indeed selected to be frequent in infants' audio recordings.

Together, these results indicate that frequency estimates based on the highest-talk segments closely approximate those derived from full daylong recordings, and that the frequency contrasts between hand-picked test nouns (audio- and video-nouns) and generic nouns are robust to the amount of audio sampled.

Table A2

Results of Kruskal-Wallis tests (nonparametric equivalent of one-way ANOVA) that compared test pair frequency across trial types (generic, audio-nouns, video-nouns) within each participant group at each month.

Participant Group	Month	Chisq	df	p-value
SEEDLingS	8	124.71	2	< .001
SEEDLingS	10	100.48	2	< .001
SEEDLingS	12	103.72	2	< .001
SEEDLingS	14	107.13	2	< .001
SEEDLingS	16	177.30	2	< .001
SEEDLingS	18	181.26	2	< .001
Control (Proxy Estimates)	8	28.72	2	< .001
Control (Proxy Estimates)	10	14.14	2	< .001
Control (Proxy Estimates)	12	9.10	2	0.011
Control (Proxy Estimates)	14	20.00	2	< .001
Control (Proxy Estimates)	16	18.56	2	< .001
Control (Proxy Estimates)	18	3.00	2	0.223

Note. The Kruskal-Wallis tests were conducted using the ‘stats::kruskal.test()’ function in R.

Table A3

Results of pairwise Wilcoxon Rank Sum Tests (nonparametric equivalent of two-sample t-tests) with Bonferroni corrections that compared test pair frequency between each combination of trial types (generic vs. audio-nouns, generic vs. video-nouns, audio- vs. video-nouns) within each participant group at each month.

Month	Generic vs. Audio-Nouns	Generic vs. Video-Nouns	Audio- vs. Video-Nouns
SEEDLingS			
8	p < .001	p < .001	p = 1
10	p < .001	p < .001	p = 0.357
12	p < .001	p < .001	p = 1
14	p < .001	p < .001	p = 1
16	p < .001	p < .001	p = 0.017
18	p < .001	p < .001	p = 0.247
Control			
8	p < .001	p = 0.525	p = 0.023
10	p = 0.111	p = 0.022	p = 0.002
12	p = 0.696	p = 0.008	p = 0.378
14	p = 1	p < .001	p = 0.004
16	p < .001	p = 1	p = 0.02
18	p = 0.192	p = 1	p = 1

Note. The pairwise Wilcoxon Rank Sum Tests were conducted using the ‘stats::pairwise.wilcox.test()’ function in R.

Table A4*Correlations between original and alternative frequency estimates.*

Temporal Window	Aggregation Level	
	Separately for Each Infant	Pooling Across All Infants
Word-Level		
Recent past (two preceding months)	Original measure	$r_p = 0.28, r_s = 0.35, b = 0.27$
Full history (all preceding months)	$r_p = 0.74, r_s = 0.77, b = 0.80$	$r_p = 0.24, r_s = 0.33, b = 0.21$
Entire corpus (all months)	$r_p = 0.62, r_s = 0.66, b = 0.69$	$r_p = 0.22, r_s = 0.30, b = 0.19$
Pair-Level		
Recent past (two preceding months)	Original measure	$r_p = 0.33, r_s = 0.39, b = 0.51$
Full history (all preceding months)	$r_p = 0.73, r_s = 0.75, b = 0.97$	$r_p = 0.28, r_s = 0.36, b = 0.42$
Entire corpus (all months)	$r_p = 0.64, r_s = 0.67, b = 0.96$	$r_p = 0.25, r_s = 0.32, b = 0.37$

Note: r_p denotes Pearson correlation among items with non-zero frequencies, r_s denotes Spearman non-parametric correlation among all items, and b denotes the fixed-effect coefficient for LogAltFreq from the model $\text{LogOriginalFreq} \sim \text{LogAltFreq} + (1|\text{Subj}) + (1|\text{Month})$ among items with non-zero frequencies.

Table A5

Fixed effects from the best-fitting model of SEEDLingS participants' word comprehension as a function of log test pair frequency (estimated based on each infant's full history up to the test month) and age in months.

Term	Estimate	SE	t	df	p-value
(Intercept)	0.065	0.018	3.62	78.25	<.001
Log Test Pair Frequency	0.122	0.030	4.03	1,641.52	<.001
Month-Scaled	0.054	0.006	9.58	1,640.74	<.001
Month-Scaled ²	0.005	0.003	1.43	1,637.79	0.152
Log Test Pair Frequency $\times$ Month-Scaled	0.031	0.013	2.50	1,631.91	0.012
Log Test Pair Frequency $\times$ Month-Scaled ²	-0.026	0.008	-3.24	1,636.23	0.001

Note. Random effects are (1|Subject).

Table A6

Fixed effects from the best-fitting model of SEEDLingS participants' word comprehension as a function of log test pair frequency (estimated based on each infant's entire corpus) and age in months.

Term	Estimate	SE	t	df	p-value
(Intercept)	0.068	0.019	3.61	88.00	<.001
Log Test Pair Frequency	0.131	0.034	3.80	1,666.38	<.001
Month-Scaled	0.056	0.006	9.39	1,661.04	<.001
Month-Scaled ²	0.004	0.004	1.10	1,659.93	0.271
Log Test Pair Frequency $\times$ Month-Scaled	0.033	0.016	2.07	1,655.75	0.039
Log Test Pair Frequency $\times$ Month-Scaled ²	-0.026	0.009	-2.74	1,658.29	0.006

Note. Random effects are (1|Subject).

Table A7

Estimated frequency effects on word comprehension at each test age, for alternative within-child frequency measures.

Frequency measure	10 months	12 months	14 months	16 months
Recent past (two preceding months)*	0.06	0.11	0.11	0.06
Full history (all preceding months)	0.06	0.12	0.13	0.08
Entire corpus (all months)	0.07	0.13	0.14	0.09

Note. Values reflect the estimated change in word comprehension (increase in target looking) associated with a one-unit increase in log-frequency at each test age. Larger positive values indicate stronger frequency effects. *Frequency computed based on the recent past is reported in the main text and listed here for comparison.

Appendix B

Exclusion Criteria

The text below provides a greater degree of detail regarding exclusions than the main manuscript for curious readers.

7.5 SEEDLingS

Our final SEEDLingS dataset included 6,165 trials from 254 infant visits ($M = 24.27/32$ trials, $SD = 6.94$, range: 8-32). During four visits, the participants had to restart the experiment due to technical errors; we retained only the most recent attempts. We originally planned for 8,448 trials in total ($44 \text{ subjects} \times 6 \text{ visits} \times 32 \text{ planned trials per visit}$). However, 60 trials were skipped as the experiment ended early due to infant fussiness/crying.

Among the remaining 8,388 planned trials, 834 were attempted twice, and 15 were attempted three times, as the experimenter deemed the initial attempts likely unusable. This resulted in 9,252 trial attempts.

From these trial attempts, we applied several exclusion criteria. First, we removed 64 trials in which participants did not hear the planned stimuli due to infant fussiness/crying, experimenter error, parental failure to say the target word, or technical difficulties (as confirmed from real-time notes and videos of experiment sessions checked post-hoc). Another 2 trials were excluded because the experimenter failed to press a button when the parent said the target word, and we could not determine the target onset (due to noisy and/or missing experiment video). Additionally, 186 trials were removed due to a complete lack of fixation data, and another 2,640 trials were excluded because participants failed to fixate on either the target or distractor for more than $1/3$ of the analysis window.

After these exclusions, we retained 6,360 trial attempts for 6,206 out of 8,448 originally planned trials. For trials with multiple attempts, we kept only the most recent

usable attempt. Finally, we excluded 10 participants who contributed too few data ($< 25\%$ of planned trials), along with an additional 41 remaining trials from these participants.

7.6 Control

Our final control dataset included 6,151 trials from 247 infant visits ($M = 24.90/32$ trials, $SD = 6.22$, range: 8-32). During 9 visits, the participants had to restart the experiment due to technical errors; we retained only the most recent attempts. Participants from another 25 visits were replaced due to fussiness ($n = 4$), technical issues ($n = 4$), experimenter error ($n = 4$), or ineligibility discovered after the session ($n = 13$). In two further visits, the participant was not replaced, and no further data were collected due to parental interference or inaudible parental speech. We originally planned for 8,448 trials in total ($44 \text{ subjects} \times 6 \text{ months} \times 32 \text{ planned trials per visit}$). After excluding data from the two unusable but unreplaced visits (64 trials), another 178 trials were skipped as the experiment ended early due to infant fussiness or technical issues.

Among the remaining 8,206 planned trials, 107 were attempted twice, and 3 were attempted three times, as the experimenter deemed the initial attempts likely unusable. This resulted in 8,383 trial attempts.

From these trial attempts, we applied the same exclusion criteria as above. First, we removed 86 trials in which participants did not hear the planned stimuli due to infant fussiness/crying, experimenter error, parental failure to say the target word, or technical difficulties (as confirmed from real-time notes and videos of experiment sessions checked post-hoc). Another 78 trials were excluded because the experimenter failed to press a button when the parent said the target word, and we could not locate the target onset (due to noisy and/or missing experiment video). Additionally, 190 trials were removed due to a complete lack of fixation data, and another 1,717 trials were excluded because participants failed to fixate on either the target or distractor for more than $1/3$ of the analysis window.

1639 After these exclusions, we retained 6,312 trial attempts for 6,204 out of 8,448
1640 originally planned trials. For trials with multiple attempts, we kept only the most recent
1641 usable attempt. Finally, we excluded 15 participants who contributed too few data ($< 25\%$
1642 of planned trials), along with their additional 53 remaining trials.

Appendix C

R Packages

We used R (Version 4.2.1; R Core Team, 2022) and the R-packages *assertr* (Version 3.0.1; Fischetti, 2023), *blabr* (Version 0.25.1; Lab, n.d.), *broom* (Version 1.0.5; Robinson et al., 2023), *dplyr* (Version 1.1.4; Wickham, François, et al., 2023), *forcats* (Version 1.0.0; Wickham, 2023), *ggh4x* (Version 0.3.1; van den Brand, 2025), *ggplot2* (Version 3.5.2; Wickham, 2016), *kableExtra* (Version 1.4.0; Zhu, 2024), *lme4* (Version 1.1.36; Bates et al., 2015), *lmerTest* (Version 3.1.3; Kuznetsova et al., 2017), *lubridate* (Version 1.9.2; Grolemund & Wickham, 2011), *MASS* (Version 7.3.60; Venables & Ripley, 2002), *Matrix* (Version 1.6.2; Bates et al., 2024), *papaja* (Version 0.1.2; Aust & Barth, 2023), *patchwork* (Version 1.3.1; Pedersen, 2025), *purrr* (Version 1.0.2; Wickham & Henry, 2023), *readr* (Version 2.1.4; Wickham, Hester, et al., 2023), *stringr* (Version 1.5.0; Wickham, 2022), *tibble* (Version 3.2.1; Müller & Wickham, 2023), *tidyr* (Version 1.3.0; Wickham, Vaughan, et al., 2023), *tidyverse* (Version 2.0.0; Wickham et al., 2019), and *tinylabels* (Version 0.2.4; Barth, 2023) for all our analyses.

Appendix D

More Details on Comparing SEEDLingS and Control Participants

This section provides further details on the statistical procedures and results comparing SEEDLingS and control participants in terms of the amount of retained data, vocabulary size, and word comprehension performance in the eyetracking experiments.

7.7 Amount of Retained Data

We first compared the two groups on the amount of retained data. If the longitudinal SEEDLingS participants became increasingly familiar with the test setup and benefited from repeated exposure, they should contribute a growing number of usable trials over time and more usable trials overall than the control group. However, Fig. 7 suggests that the two groups contributed a similar number of usable trials. This pattern was confirmed by a linear regression model predicting the number of usable trials per participant as a function of group and age. Results showed a significant main effect of age ($b = 1.14$, $SE = 0.16$, $t(498) = 6.94$, $p < .001$), indicating that both groups contributed more usable data as they got older, but the effect of group was not significant ($b_{\text{SEEDLingS}} = -0.65$, $SE = 0.56$, $t(498) = -1.15$, $p = 0.25$). Thus, we find no evidence that SEEDLingS participants contributed more usable trials than the control group, suggesting a limited role for repeated testing experience in data retention.

7.8 Vocabulary Size

We next compared the two groups on receptive and productive vocabulary sizes, as measured by the CDI. A linear model predicting CDI comprehension scores as a function of age, participant group, and their interaction revealed a significant main effect of age ($b = 37.85$, $SE = 2.87$, $t(462) = 13.18$, $p < .001$), group ($b_{\text{SEEDLingS}} = 47.78$, $SE = 7.03$, $t(462) = 6.80$, $p < .001$), and a significant age $\times$ group interaction ($b_{\text{SEEDLingS:age}} = 13.94$, $SE = 4$, $t(462) = 3.48$, $p < .001$). Specifically, according to the model's fixed effects, SEEDLingS participants were reported to comprehend 48 more words than controls at 12 months.

Receptive vocabulary grew with age for both groups: Controls gained about 38 words every two months, while SEEDLingS gained an additional 14 words, widening the gap over time.

In contrast, a similar linear model predicting CDI production scores revealed only a significant main effect of age ($b = 11.20$, $SE = 1.57$, $t(462) = 7.14$, $p < .001$), but not group ($b_{\text{SEEDLingS}} = 2.42$, $SE = 3.84$, $t(462) = 0.63$, $p = 0.53$), and no interaction ($b_{\text{SEEDLingS:age}} = 2.78$, $SE = 2.19$, $t(462) = 1.27$, $p = 0.20$). This suggests that for every two-month increase in age, both SEEDLingS and control participants were reported to produce 11 more words.

However, we interpret the group difference in CDI scores with caution for two main reasons. First, the difference may reflect variation in how parents completed the surveys. For parents of longitudinal SEEDLingS participants, all CDI surveys following the initial administration were pre-filled with responses from the previous form. In contrast, parents of cross-sectional control participants completed the survey only once, without any pre-filled responses. This pre-filling may have inflated CDI scores for SEEDLingS participants, as parents may be less likely to revise or remove previously endorsed words. Second, parents participating in the yearlong SEEDLingS study may have become more attuned to their children's vocabulary over time. Their extended involvement in a word learning study could make them more sensitive to their child's word knowledge and thus more accurate in reporting word comprehension. Supporting this possibility, we observed a stronger correlation between SEEDLingS participants' CDI comprehension scores and their eyetracking performance (0.48, 95% CI = [0.38, 0.57]) than between control participants' CDI comprehension scores and eyetracking performance (0.34, 95% CI = [0.22, 0.45]). However, the confidence intervals overlap, so this difference should be interpreted cautiously.

7.9 Word Comprehension

In the main text, we reported group differences in receptive vocabulary size (CDI comprehension) and word comprehension performance across the three trial types (generic,

audio-nouns, video-nouns) in eyetracking assessment. Here, we ask whether the observed group differences in word comprehension could be attributed to differences in receptive vocabulary. To test this, we fit linear mixed-effects models predicting word comprehension performance at 18 months, where group differences appeared largest, as a function of participant group, and then added CDI comprehension as a covariate to assess whether group effect remained. A caveat is that CDI data were only available for a subset of participants, so these analyses were conducted on a reduced sample (36 out of 44 SEEDLingS participants and 34 out of 41 control participants retained for eyetracking analyses at 18 months).

7.9.1 *Generic Items*

In the reduced sample with CDI data, we first fit a linear mixed-effects model predicting the increase in target looking for generic items at 18 months as a function of group. Model selection removed the fixed effect of group and returned an intercept-only model, suggesting that the originally observed group difference may have been weak or unstable in this subset.

We then fit a second model including both group and CDI comprehension as fixed effects. Model selection again dropped the group effect but retained a significant main effect of CDI comprehension ($b = 5.87\text{e-}04$, $SE = 2.25\text{e-}04$, $t(67.70) = 2.61$, $p = 0.01$), suggesting that for every 100-word increase in receptive vocabulary size, comprehension of generic words (operationalized as increase in target looking) would increase by 0.06 unit.

In sum, the group difference in comprehension of generic items was not robust. Among participants with CDI data, individual differences in receptive vocabulary better explained comprehension of generic items than participant group.

7.9.2 *Audio-Nouns*

Using the reduced sample, we first fit a linear mixed-effects model predicting the increase in target looking for audio-nouns at 18 months as a function of group. Model

selection removed the group effect and returned an intercept-only model, suggesting that the original group difference may not be robust within this smaller sample. We then fit a second model including both group and CDI comprehension as fixed effects. Model selection removed both the group effect and the effect of CDI comprehension, resulting in another intercept-only model. Thus, the group difference in audio-noun comprehension was not robust, and receptive vocabulary did not reliably predict performance on these items either.

7.9.3 *Video-Nouns*

For video-nouns, we first fit a linear mixed-effects model predicting the increase in target looking for video-nouns at 18 months as a function of group. Model selection retained the full model, and the group effect remained significant in this reduced sample ($b_{\text{SEEDLingS}} = 0.18$, $SE = 0.06$, $t(874.50) = 3.25$, $p < .01$). We then fit a second model including both group and CDI comprehension as fixed effects. This time, model selection removed the main effect of CDI comprehension but retained the group effect, yielding the same model structure as before. This indicates that the group difference in video-noun comprehension is robust and cannot be fully accounted for by differences in receptive vocabulary.

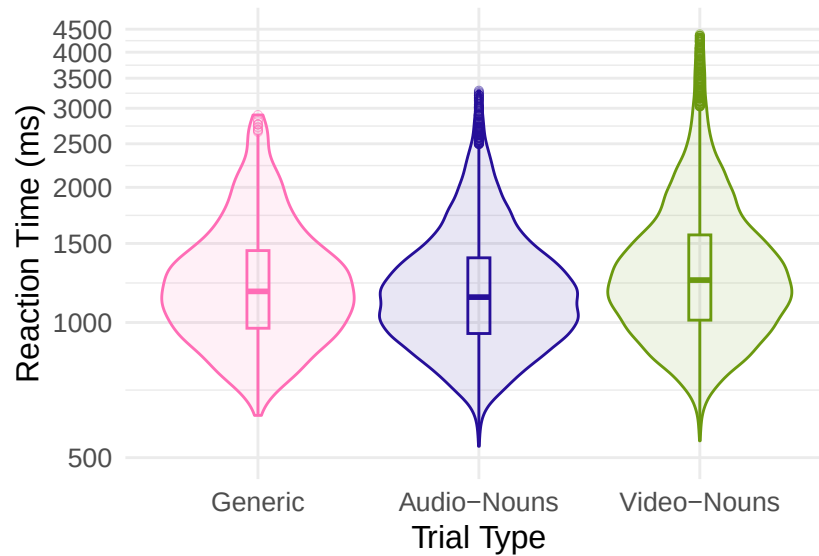
Appendix E

Adult Control Study

In order to assess how readily recognizable the hand-picked referent images (video-nouns) vs. prototypical images (audio-nouns and generic nouns) were to individuals with mature lexicons, we conducted an adult control study using the infant stimuli and a written prompt directing them to the target. A total of 113 native English-speaking adults were recruited via Prolific and compensated at an hourly rate of \$15. An additional 2 participants were excluded for failing one or more attention checks, and another 6 were excluded for providing responses that indicated they were nonnative English speakers, based on an open-ended picture description task. Each session took an average of 9.01 minutes ($SD = 4.14$, range: 4.03-25.37).

Each participant completed half of the hand-picked trials (i.e., stimuli from 22 of the 44 SEEDLingS infants) and all of the generic trials from a specific infant test month (8, 10, 12, 14, 16, 18 months). This resulted in a total of 184 trials per participant: 8 generic, 88 audio-noun, and 88 video-noun trials. In each trial, participants saw yoked image pairs along with a prompt sentence (e.g., “Can you find the X?”) and were instructed to click on the image corresponding to the labeled referent, X. We measured participants’ reaction times (RTs) in making their selections. If the hand-picked referents in the video-noun trials were less recognizable than the prototypical referents in the audio-noun and generic trials, we would expect longer reaction times in the video-noun trials. We excluded 275 trials where RTs were more than 3 standard deviations from the mean for that trial type. These exclusions constituted <2% of test trials overall and within each trial type. Because RTs were heavily skewed, we applied a log transformation in subsequent statistical analyses. See Fig. E1 for a visualization of the RT distribution by trial type and Fig. E2 for RTs of the most extreme RT values on average in each condition.

To test whether RTs differed between trial types, and whether any observed

**Figure E1**

Adult reaction time distribution by trial type (generic, audio-nouns, video-nouns) across all infant test months. Boxes indicate the interquartile range and horizontal lines indicate medians. Empty dots indicate potential outliers with extreme reaction times. Adults were significantly slower to respond in video-noun trials than audio-noun and generic trials. Reaction times in audio-noun and generic trials were similar.

1773 differences were consistent across infant test months, we began by fitting the following full
 1774 linear mixed-effects model: $\log RT \sim \text{month} \times \text{trial type} + (1|\text{subject}) + (1|\text{test}$
 1775 $\text{pair})$. Month was encoded as a categorical variable, as we were comparing between
 1776 months rather than tracking developmental change over time. Trial type was dummy coded
 1777 with video-nouns as the reference level, since our primary interest was in contrasting the
 1778 hand-picked referents in video-nouns trials with the prototypical referents in audio-nouns
 1779 and generic trials. Model selection dropped the interaction and the main effect of month
 1780 but retained trial type, indicating that the adults processed different types of stimuli at
 1781 different speeds, and that this effect was consistent across all infant test months (final
 1782 model: $\log RT \sim \text{trial type} + (1|\text{subject}) + (1|\text{test pair})$). Results showed that
 1783 video-noun trials took significantly longer for adults to process than both audio-nouns
 1784 ($b_{\text{Audio-Nouns}} = -0.04$, $SE = 0.007$, $t(1,060.40) = -18.14$, $p < .001$) and generic trials

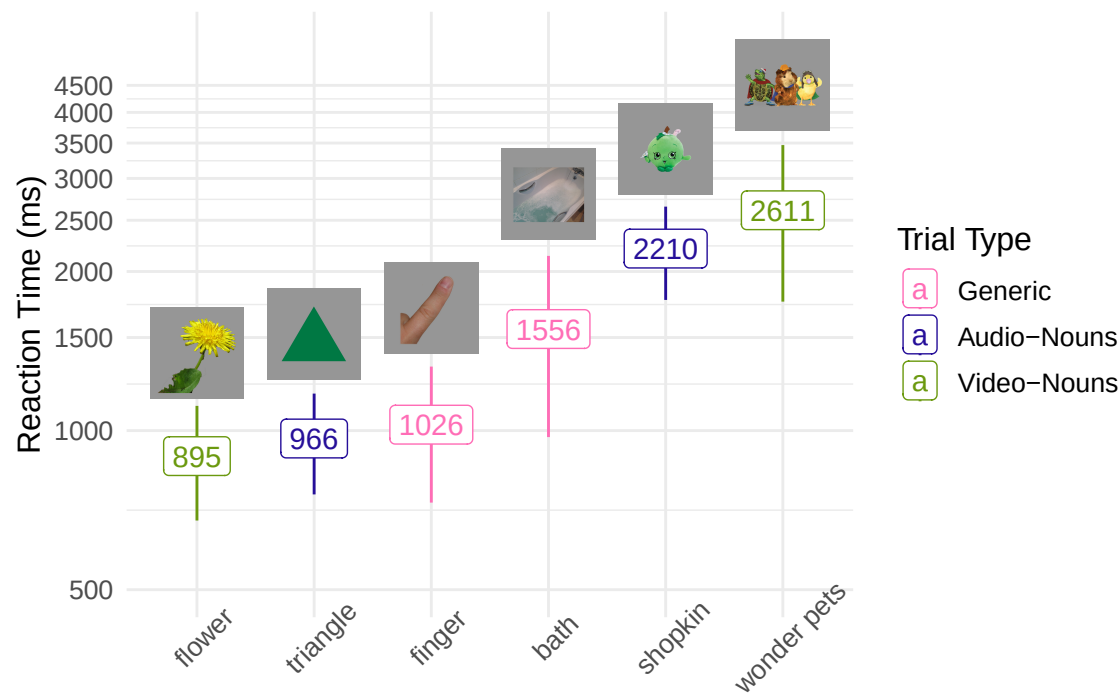


Figure E2

Examples of referents with the fastest and slowest adult reaction times by trial type (generic, audio-nouns, video-nouns). Numbers in each box indicate mean reaction times in milliseconds, and vertical lines represent standard deviations.

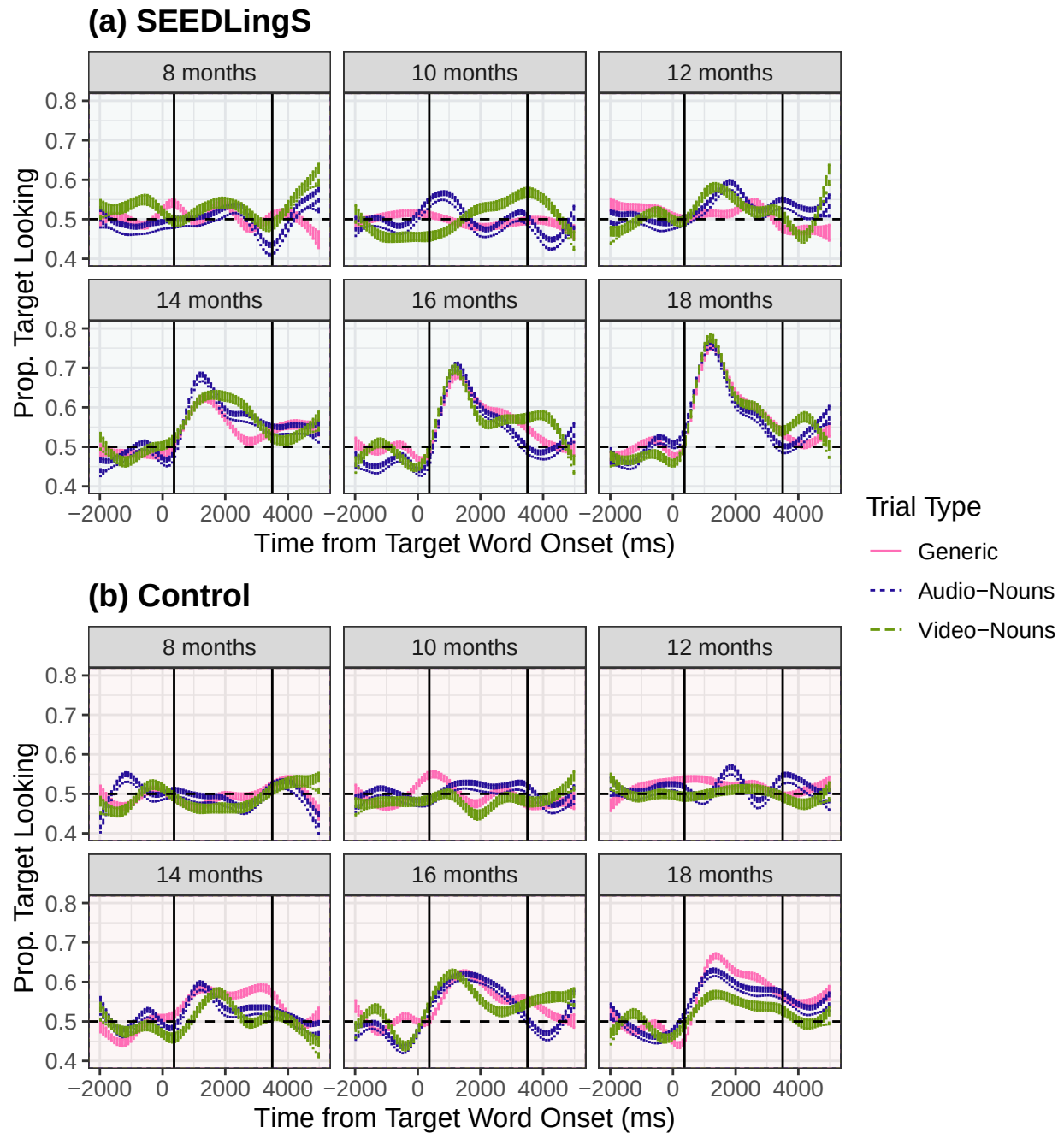
1785 ($b_{\text{Generic}} = -0.03$, $SE = 0.007$, $t(739.63) = -3.75$, $p < .001$).

1786 Overall, consistent with our prediction, it took adults significantly longer to select
1787 the correct referents in video-noun trials than in audio-noun or generic trials. This suggests
1788 that the difficulties observed in control infants on video-nouns may stem from properties of
1789 referents (e.g., low typicality) rather than overly narrow semantic representations.

Appendix F

Timecourse Visualizations

1790 To complement our primary analyses, we provide two sets of timecourse visualizations that
1791 offer a finer-grained look at infants' gaze behavior during the eyetracking experiments. The
1792 first set (Fig. F1) displays the proportion of target looking over trial time by age and trial
1793 type. The second set (Fig. F2) presents onset-contingent switch plots, which captures
1794 infants' gaze shifts between the target and distractor images as a function of their starting
1795 position.

**Figure F1**

Time course of proportion of target looking among (a) SEEDLingS (top) and (b) control participants (bottom), split by age in months and trial type (generic, audio-nouns, video-nouns). Target word onset (e.g., “kitty” in “Do you see the kitty?”) is marked as Time 0. Vertical lines indicate the analysis window (367–3,500 ms). The dashed horizontal line represents chance-level looking (0.5).

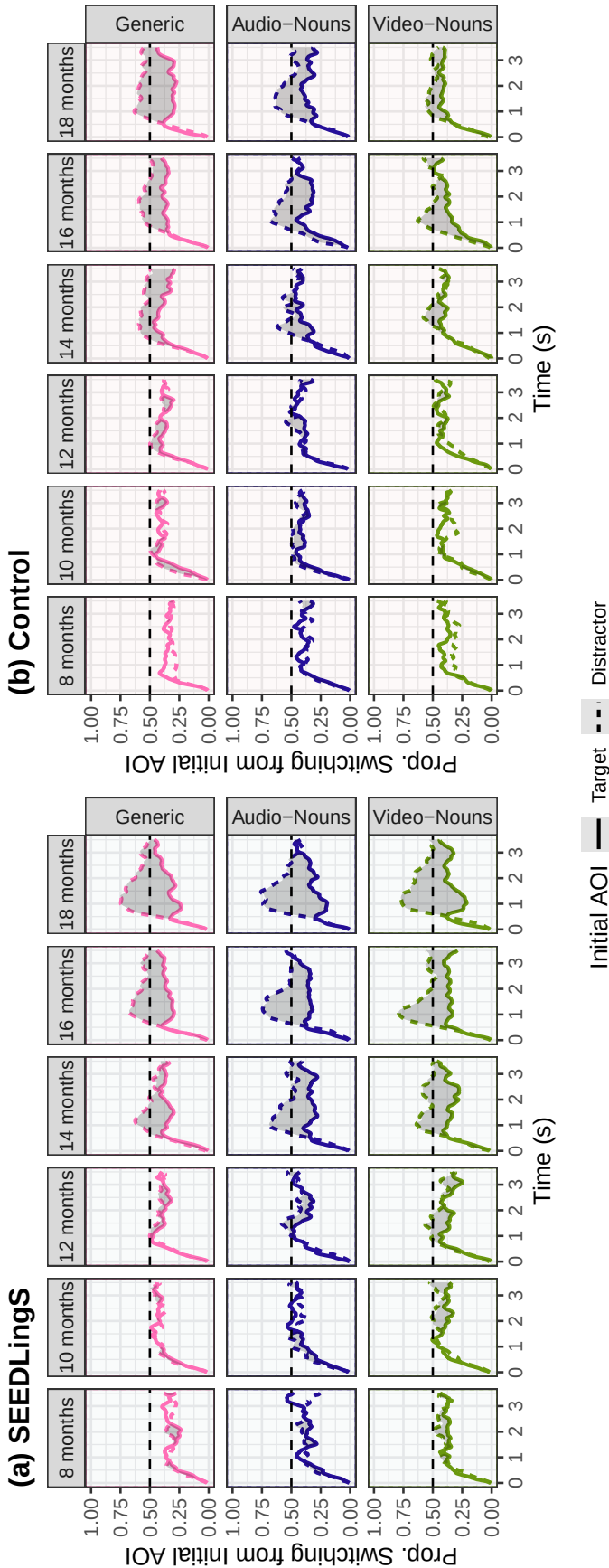


Figure F2

Onset-contingent plots comparing different types of test items (generic, audio-nouns, video-nouns) among (a) SEEDLingS (top) and (b) control participants (bottom). Each panel displays the proportion of shifts from the initial AOIs to the other over time, first averaged within participants across trials, then across participants. The solid line represents children who began on the target, and the dotted line represents children who began on the distractor. The gray shaded region highlights periods where more switches occurred from distractor to target than from target to distractor, reflecting how well children discriminated between the two test items based on the provided label. A larger gray region indicates better performance. For SEEDLingS participants, the size of the gray region appears consistent across test item types over months, suggesting that familiarity plays a small role in word recognition. In contrast, for control participants, the size of the gray region appears comparable between generic and audio-nouns, but the gap between these two and video-nouns widened over months, with video-nouns showing the poorest performance. This suggests that visual unfamiliarity may impede word recognition later in development.